Embedded System Design

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innovate achieve lead

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AEZG512/AELZG512/DMZG512
Embedded System Design
Remote Lab Contact Session-02

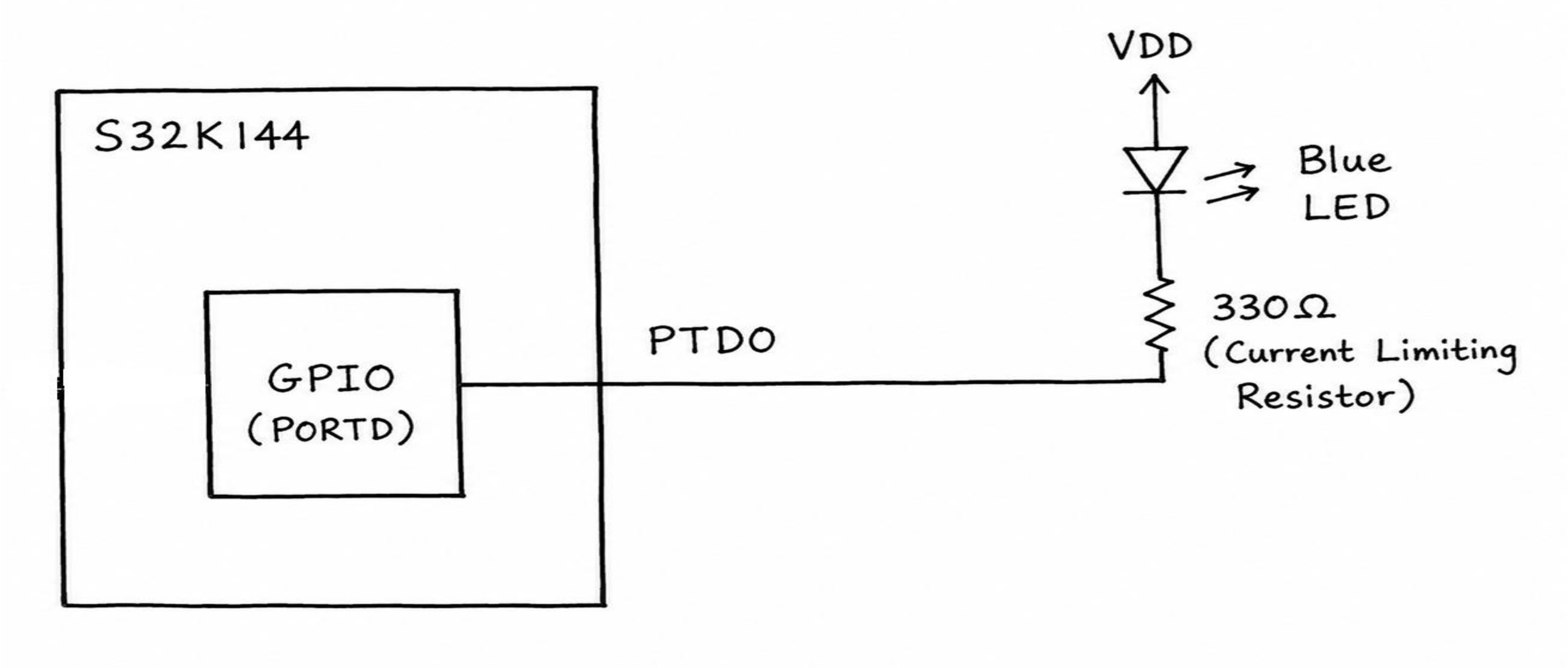
Contents

- Expt.-1: Implementation of Digital Output Control Using GPIO to Blink an LED on the S32K144 Microcontroller
- Expt.-2: Implementation of Delay Generation Using Timer (LPTMR) on the S32K144 Microcontroller

Expt.-1: Implementation of Digital Output Control Using GPIO to Blink an LED on the S32K144 Microcontroller board.

- **Objective**
  - ✓ Develop an Embedded C application using S32 SDK GPIO / Pins driver to configure a digital output pin (PTD0/PTD15/PTD16) of the S32K144 microcontroller and blink an LED at a visible rate.
- **Problem Description**
  - ✓ The S32K144 microcontroller must control onboard LED using a GPIO pin (PTD0/PTD15/PTD16). The selected GPIO pin shall be configured as a digital output. The application shall toggle the LED continuously using a software delay or SDK delay function so that the ON and OFF states are clearly visible.
- **Functional Requirements**
  - Configure the required system clock using SDK clock configuration.
  - Enable and configure the selected LED pin as GPIO output.
  - Initialize the pin state to a known logic level.
  - Toggle the LED continuously inside an infinite loop.
  - Use a delay to make blinking visible to the human eye.
  - Verify the output using LED observation.
- **Expected Output**
  - The LED connected to the selected GPIO pin should blink continuously with a fixed ON/OFF interval.

- Refer the **Project Creation in S32 Design Studio** from S1_26_AE_AEL_DM_ZG512_Lab_CS_01.pdf



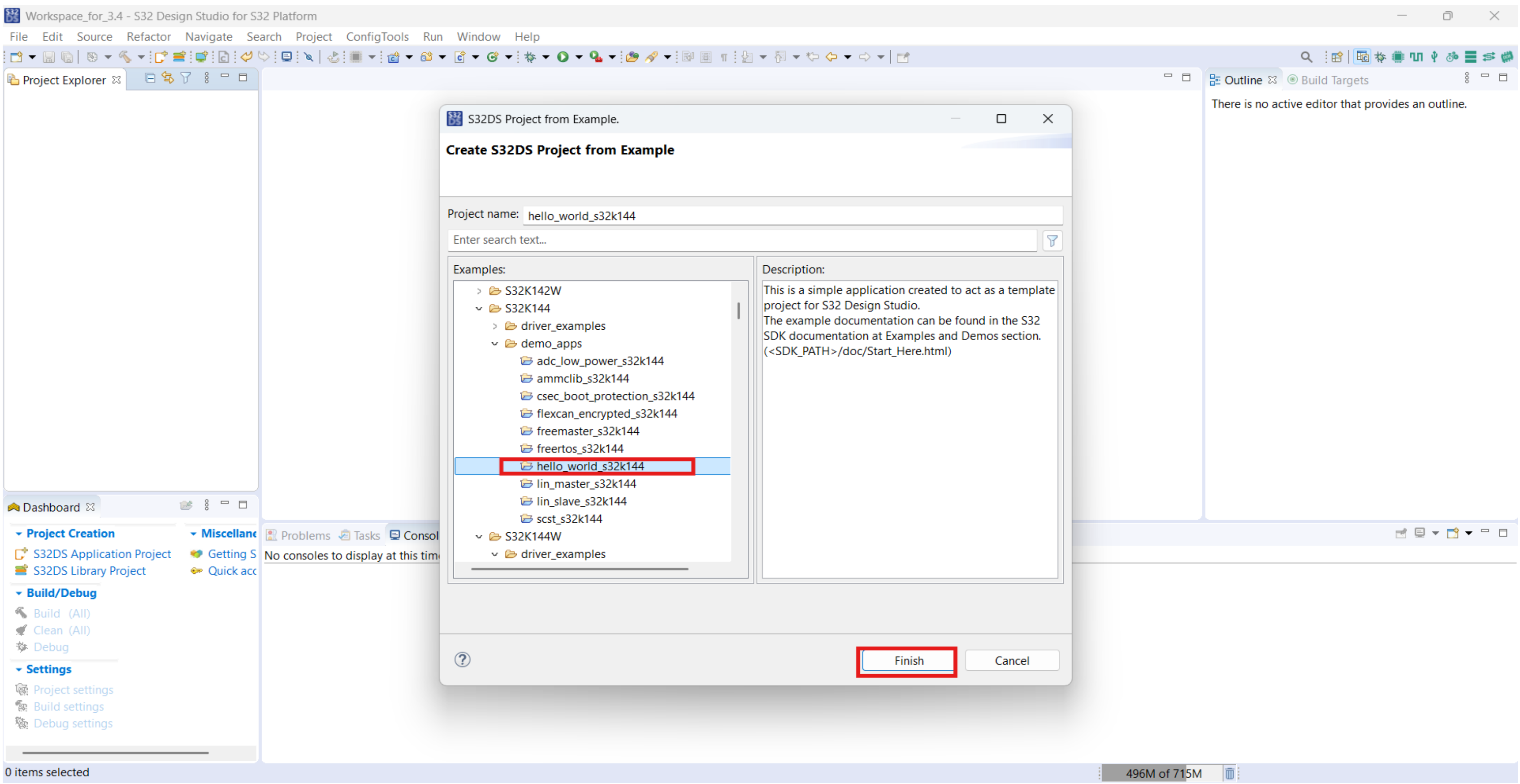
Expt.-2: Implementation of Delay Generation Using Timer (LPTMR) on the S32K144 Microcontroller.

- **Objective**
  - ✓ Develop an Embedded C application using S32 SDK timer driver to generate an accurate delay using LPTMR module instead of a software busy-wait loop.
- **Problem Description**
  - ✓ Configure a hardware timer module LPTMR to generate a fixed delay. Use this delay to toggle an LED or GPIO output (PTD0/PTD15/PTD16) at a known interval.
- **Functional Requirements**
  - Configure system clock and peripheral clock using SDK configuration tools.
  - Select LPTMR as the hardware timer source.
  - Configure the timer to generate a delay of a specified duration, for example 500 ms or 1 second.
  - Poll the timer status flag or use SDK blocking delay functionality.
  - Toggle a GPIO output after each timer delay.
  - Compare timer-based delay with software delay.
- **Expected Output**
  - The GPIO pin or LED should toggle at a stable delay interval generated by LPTMR.

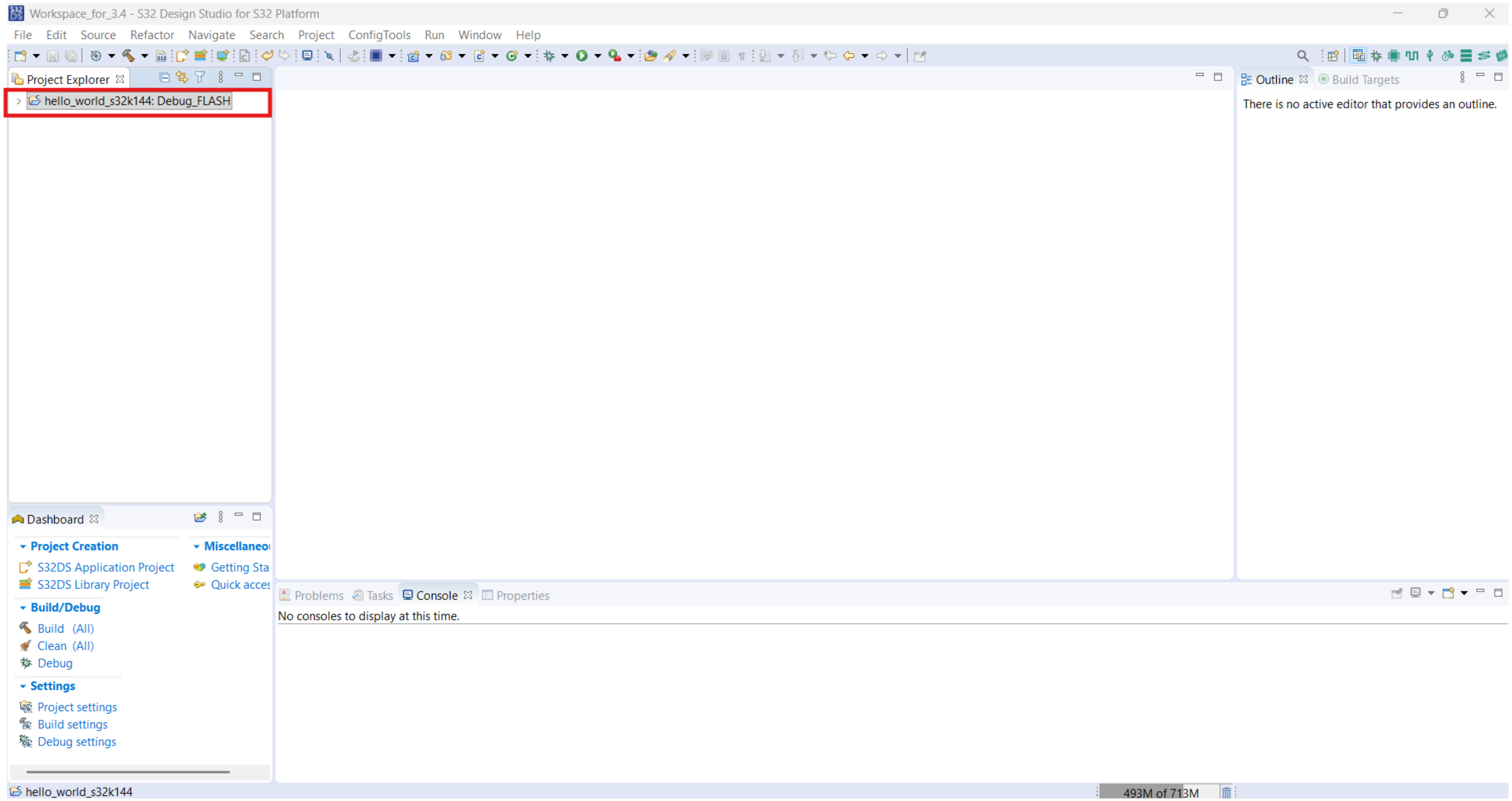
Applications of LPTMR

- Ultra-Low-Power Wake-Up Timing: Serving as a periodic system heartbeat to wake the CPU from deep sleep modes (e.g., VLPS) at defined intervals to sample sensors while keeping active power draw minimal.
- Asynchronous Pulse / Event Counting: Tracking external asynchronous pulses (such as flow meter contact closures, utility meter rotations, or button pushes) without waking the MCU core.
- ADC / Peripheral Hardware Pre-Triggering: Generating cyclic trigger strobes via TRGMUX to initiate periodic ADC conversions or comparator round-robin sequences in low-power modes.

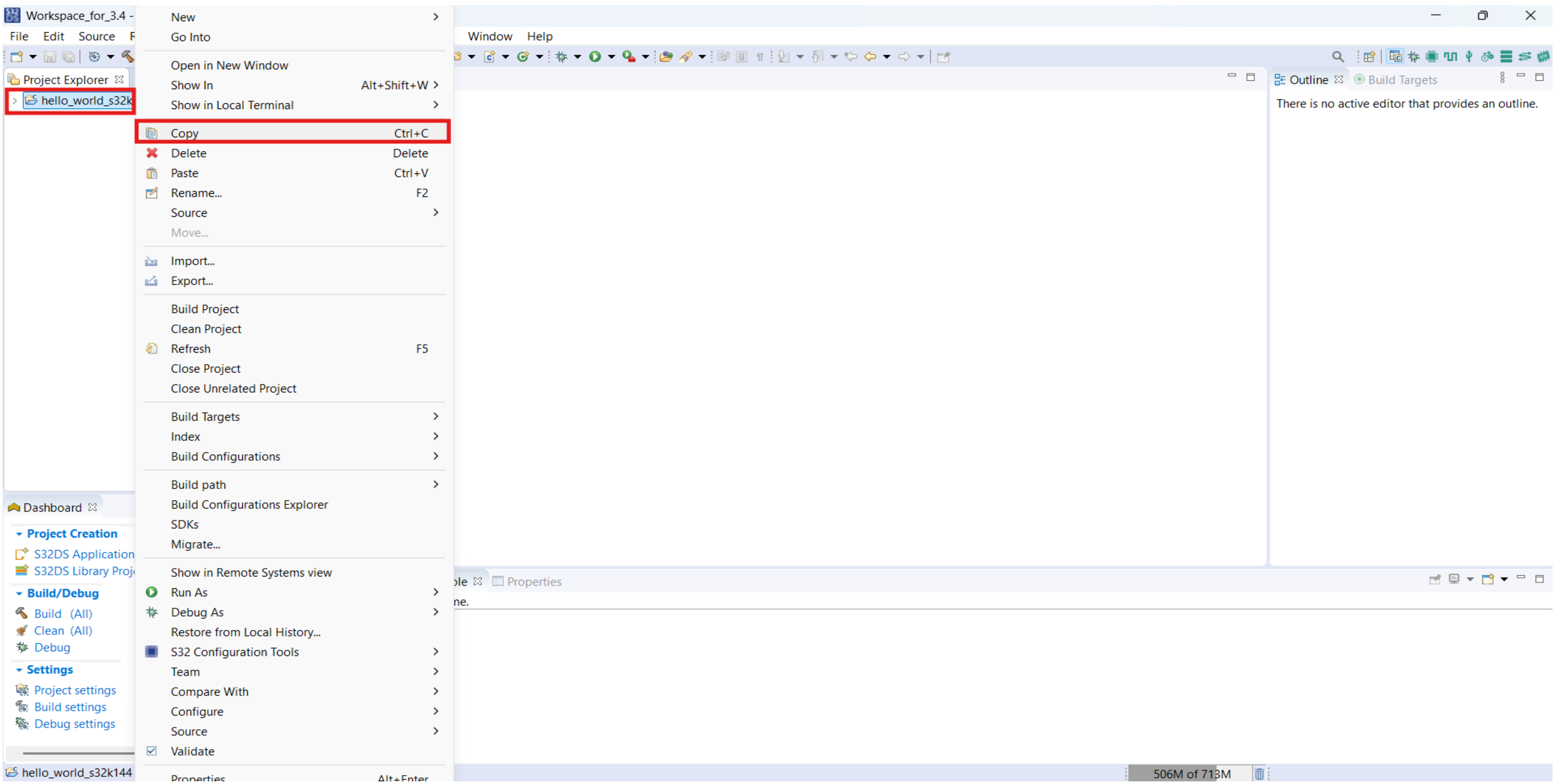
Select hello_world_s32k144 example and click on Finish. This Project blinks the on board RGB LED which is attached at pin PD15 and PD16 of the S32K144 MCU



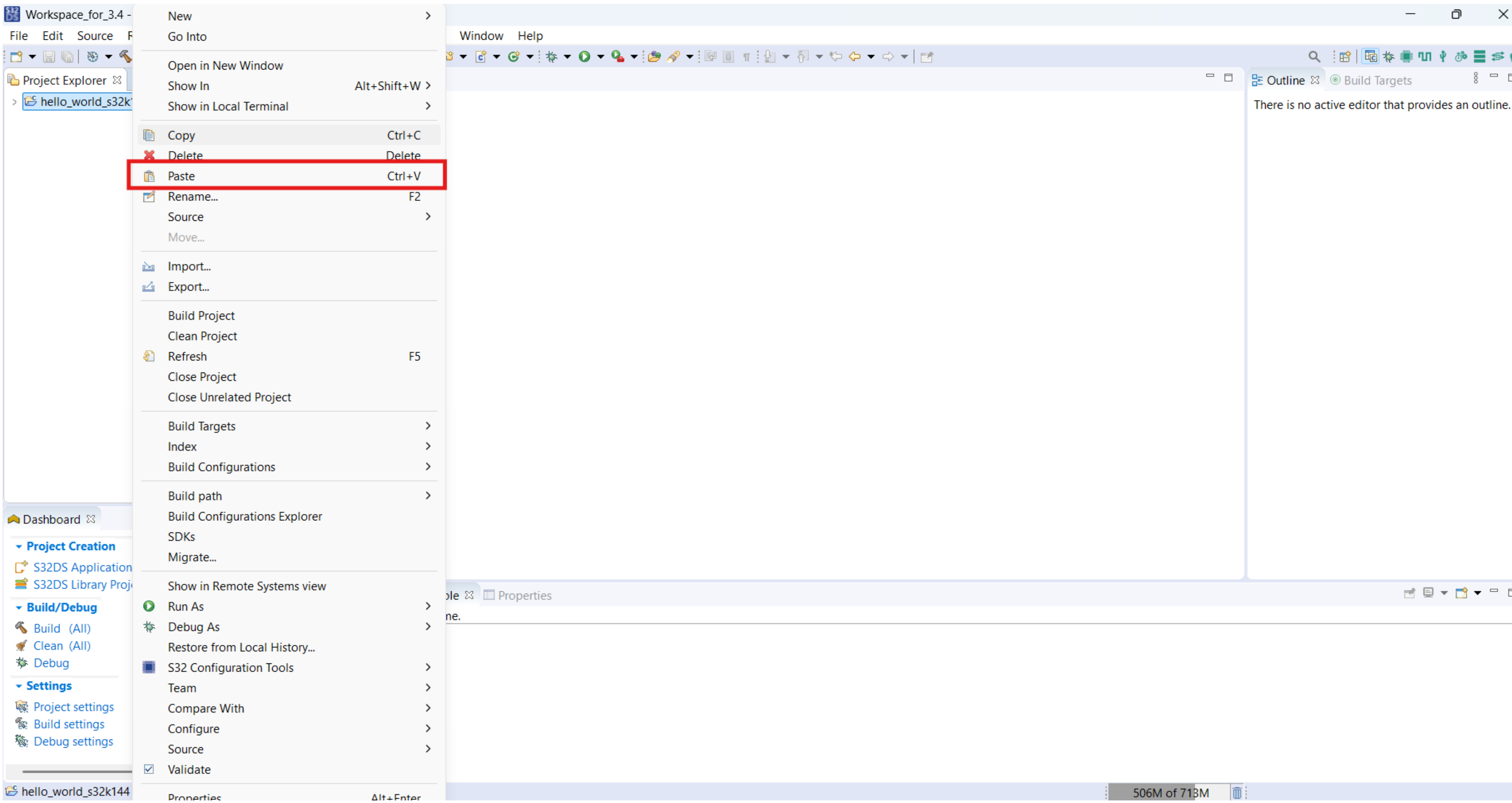
- Project is available in Project Explorer



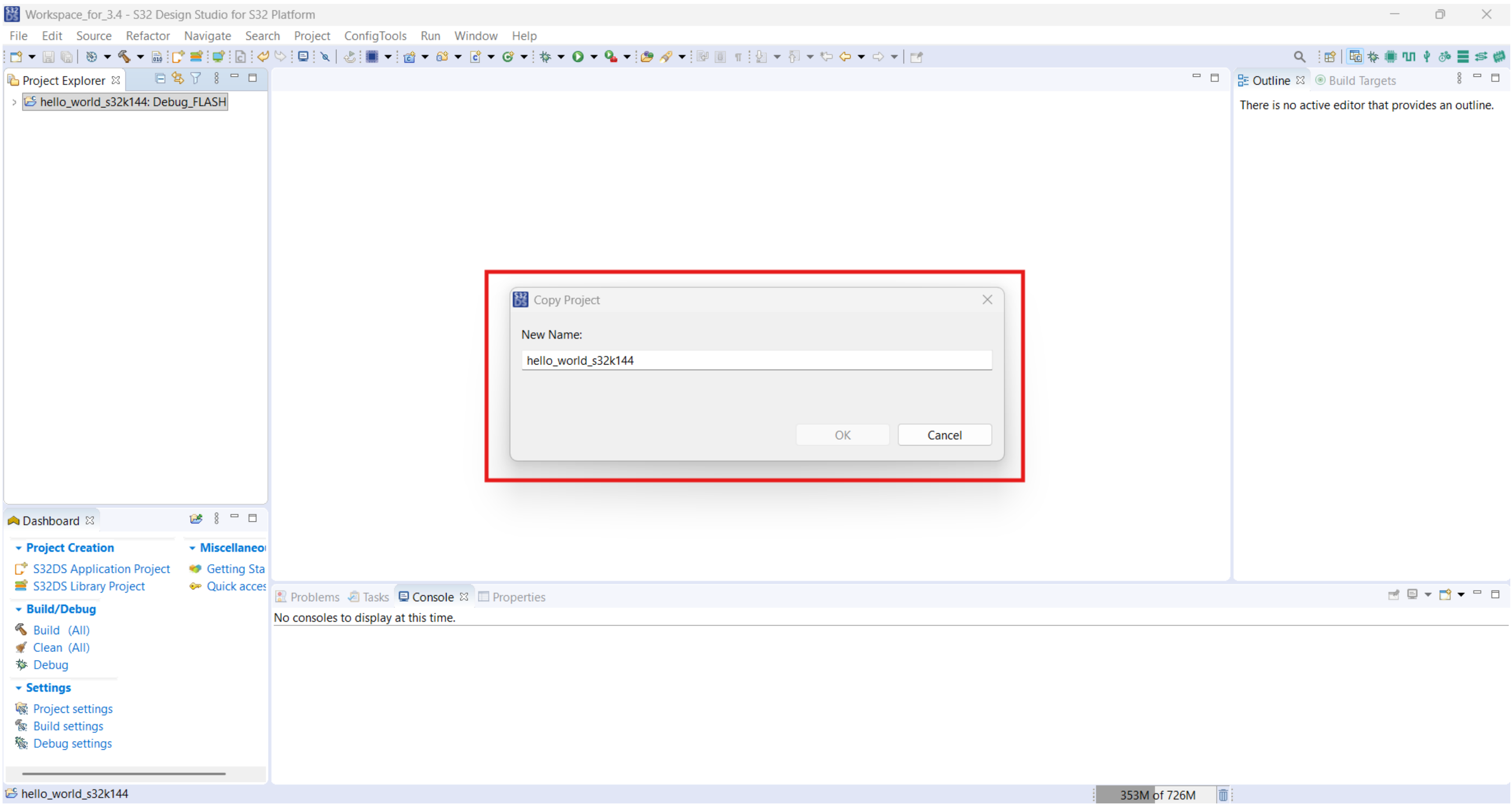
- Select hello_world_s32k144 project and right click and select **Copy**



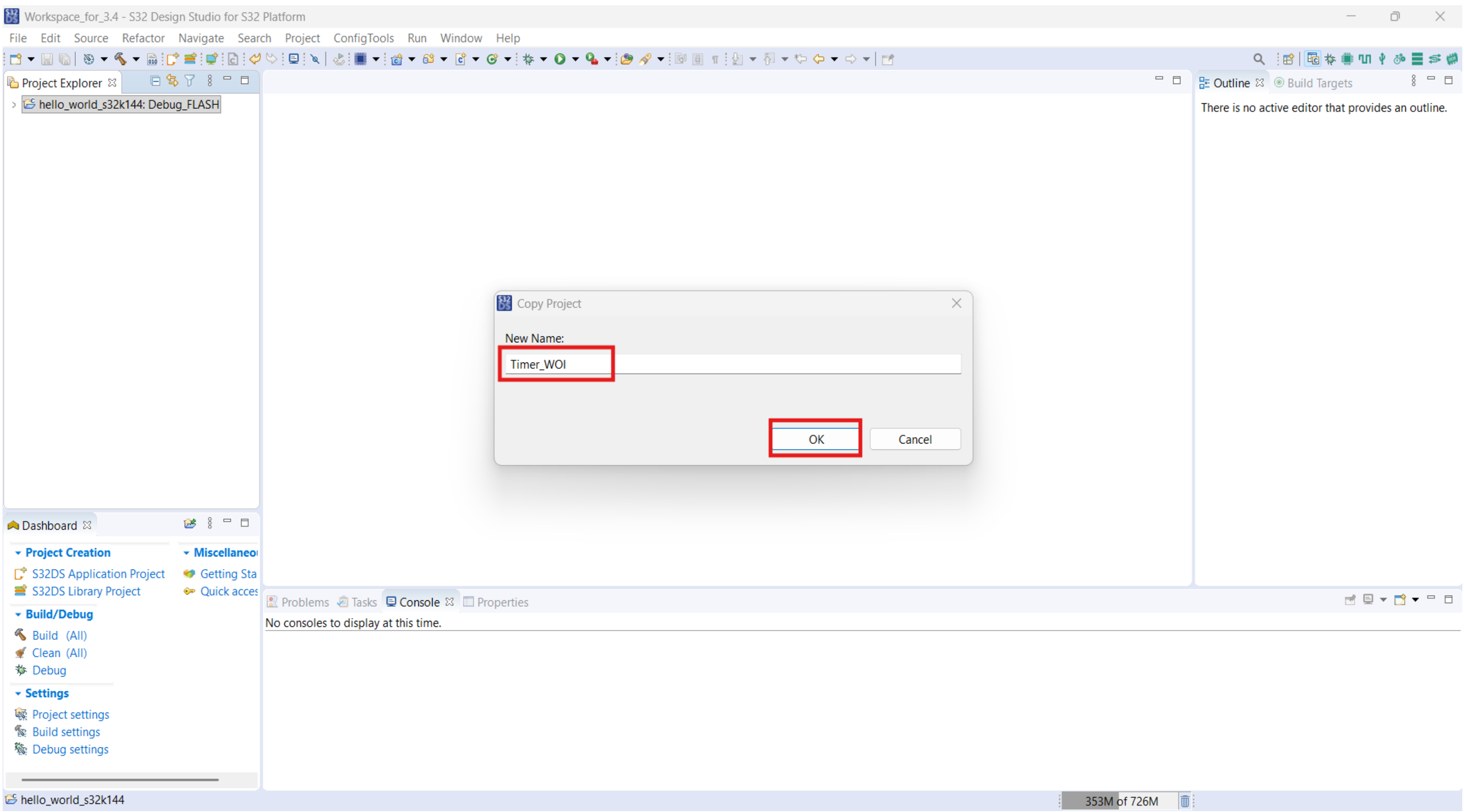
- Now click on **Paste**



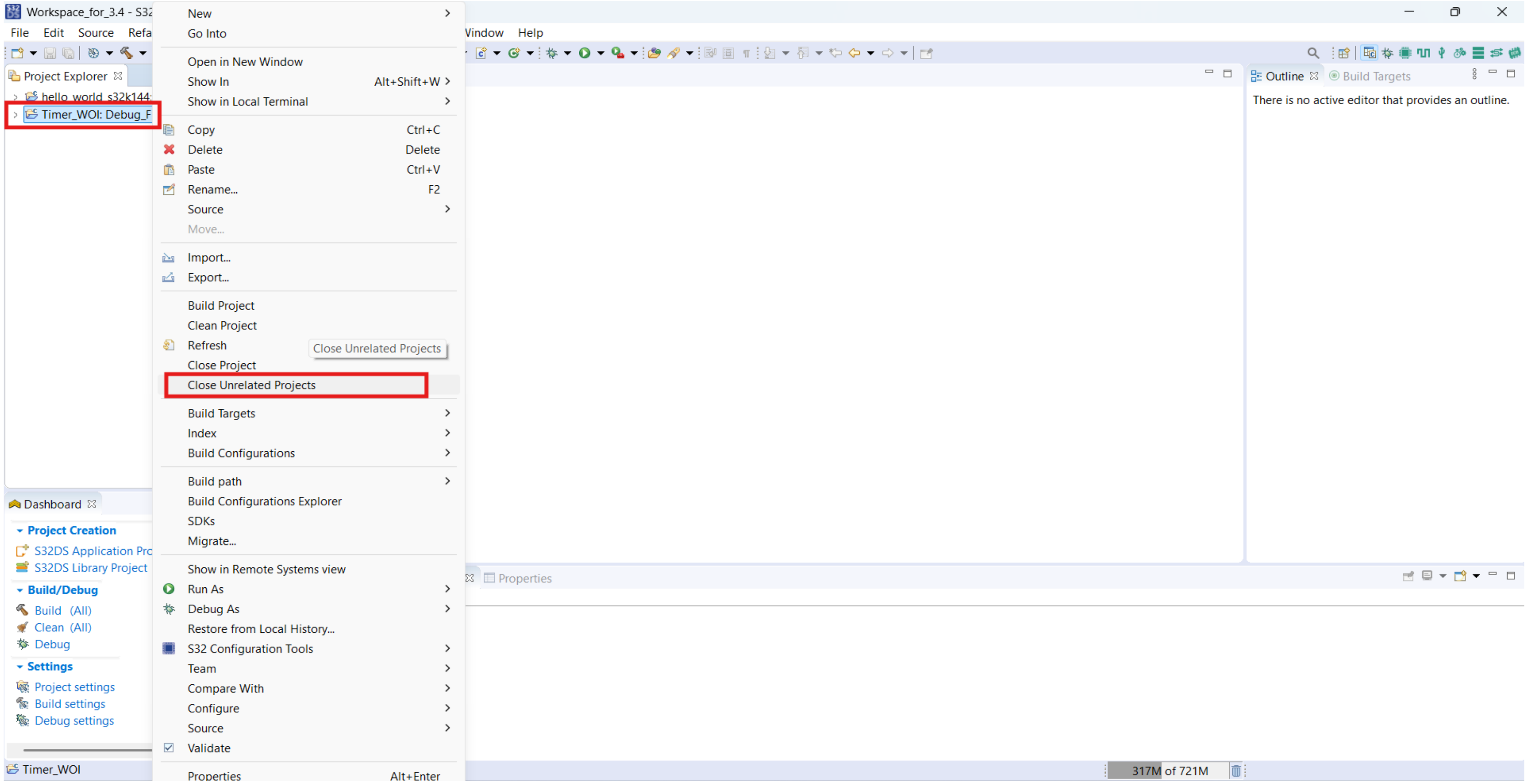
- Pop up window opens



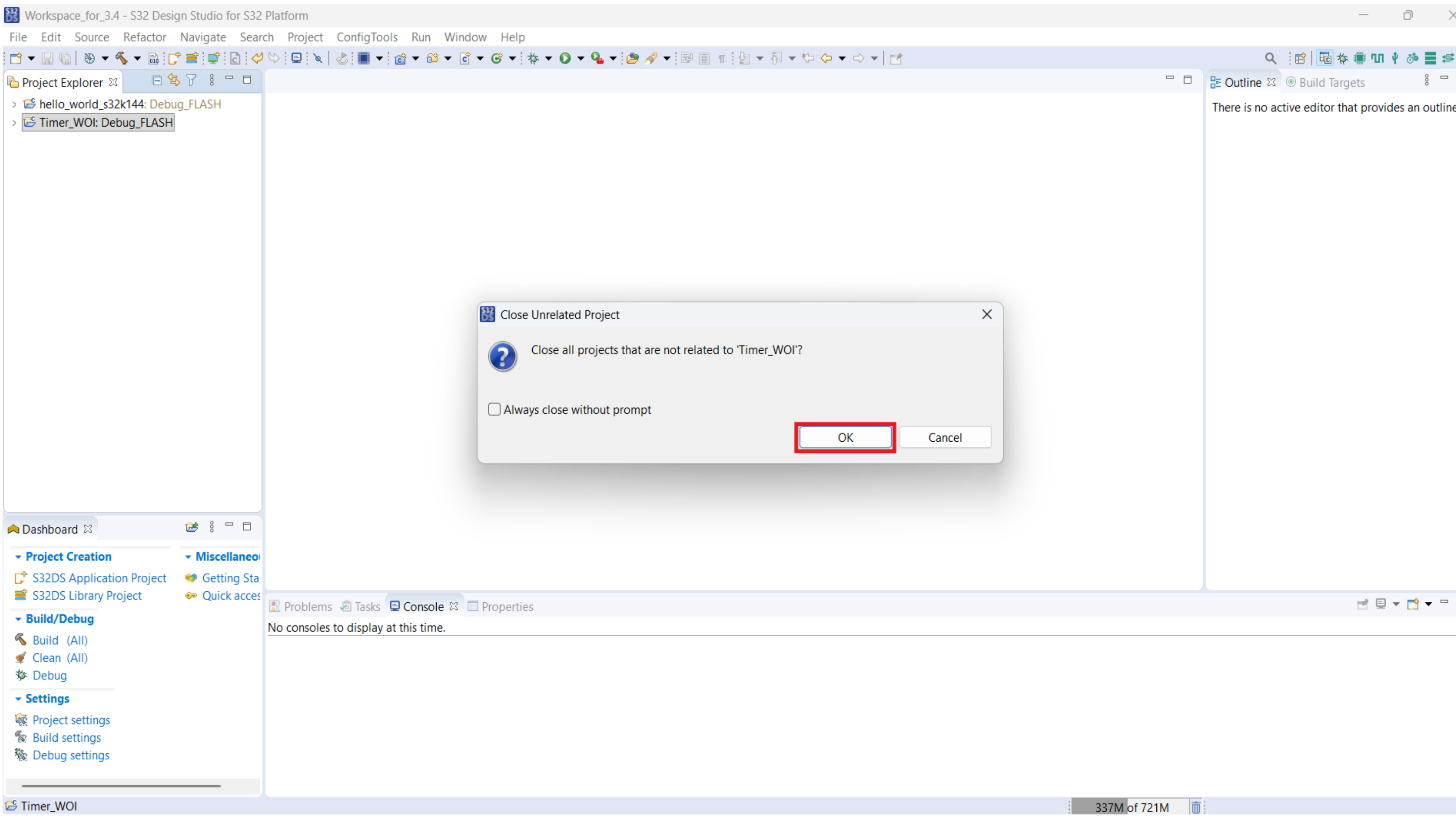
- Give name to project as Timer_WOI and Click on OK

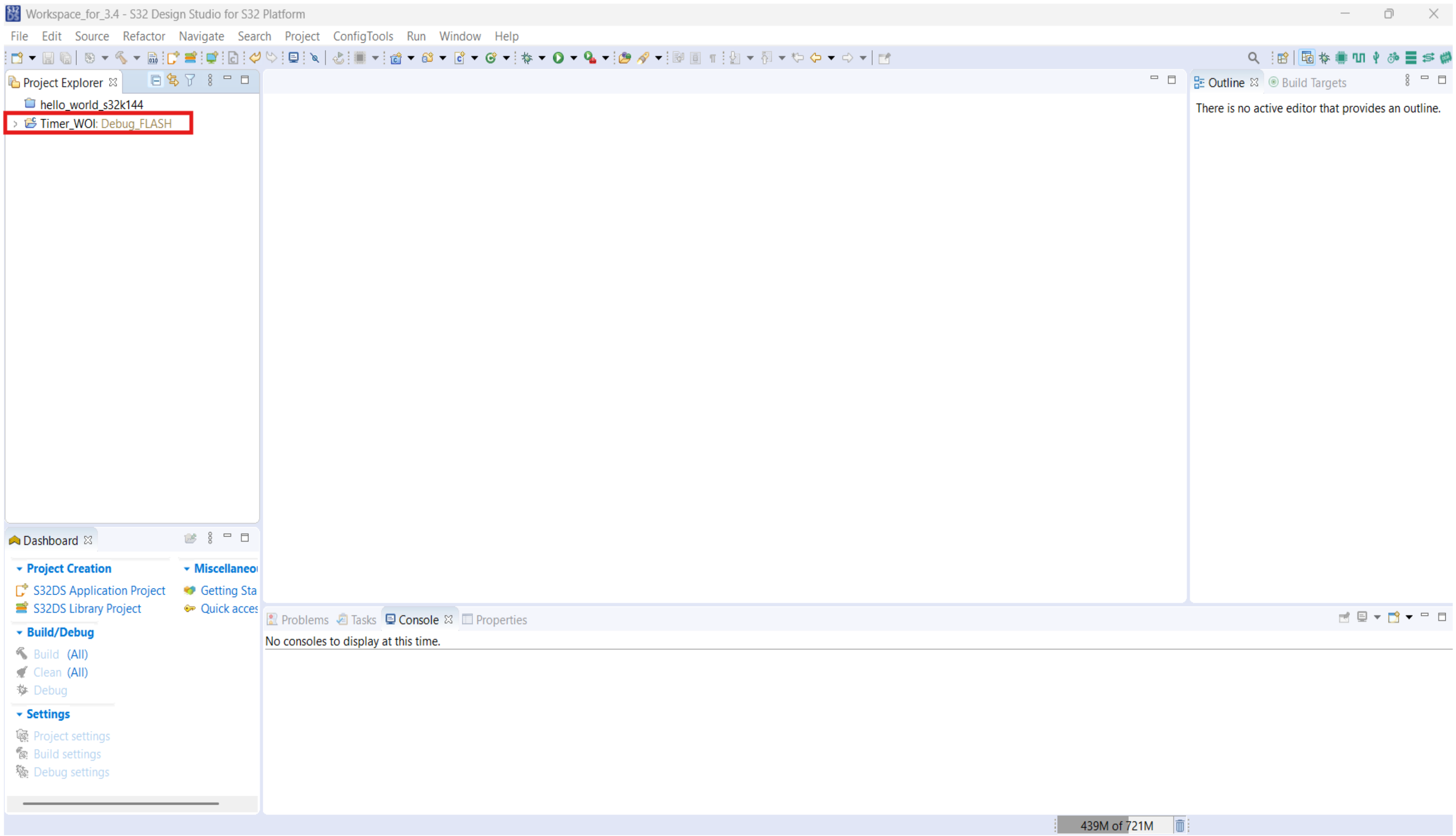


- Select Timer_WOI project and right click and click on **Close Unrelated Projects**

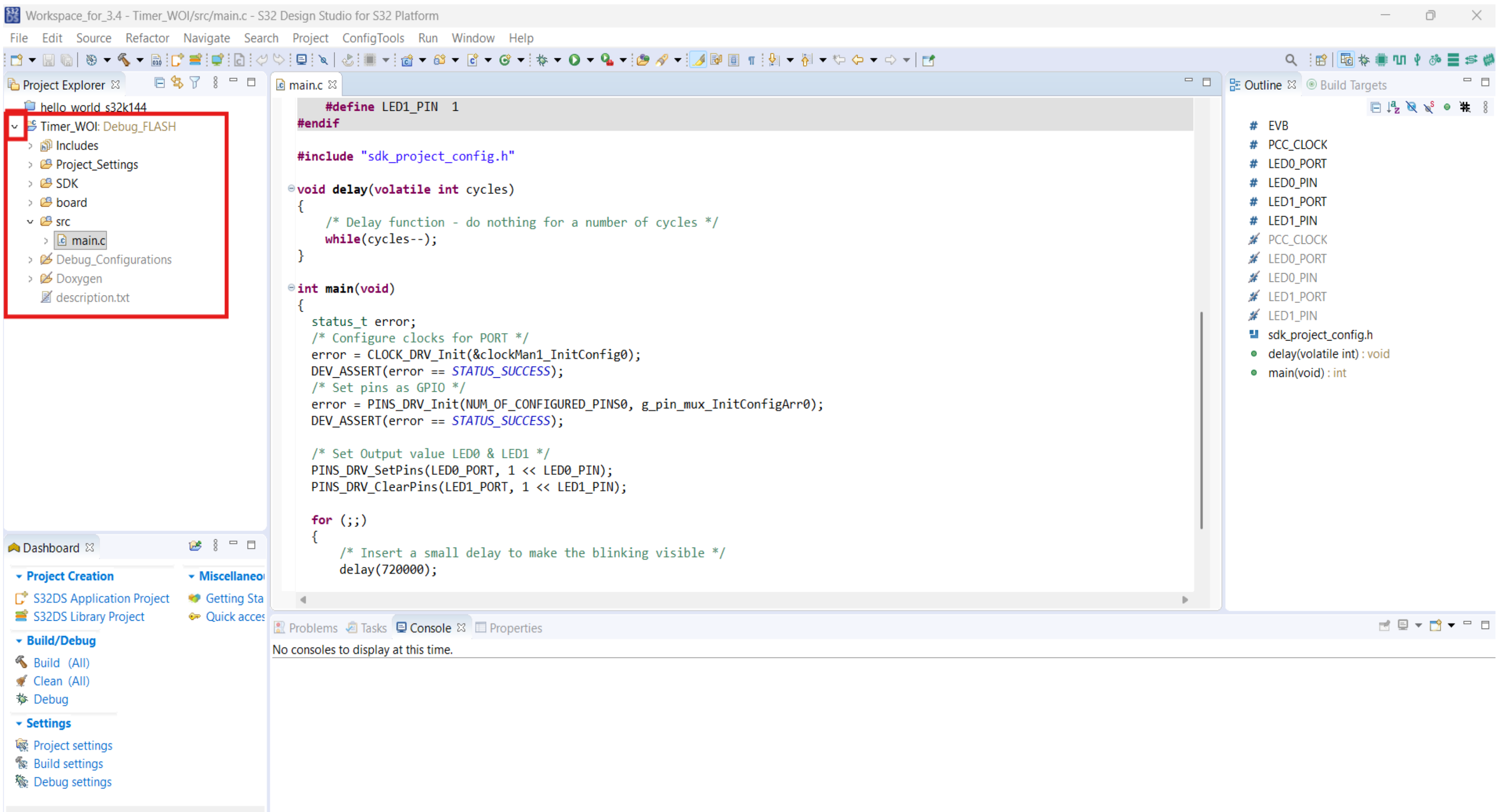


- Click on OK

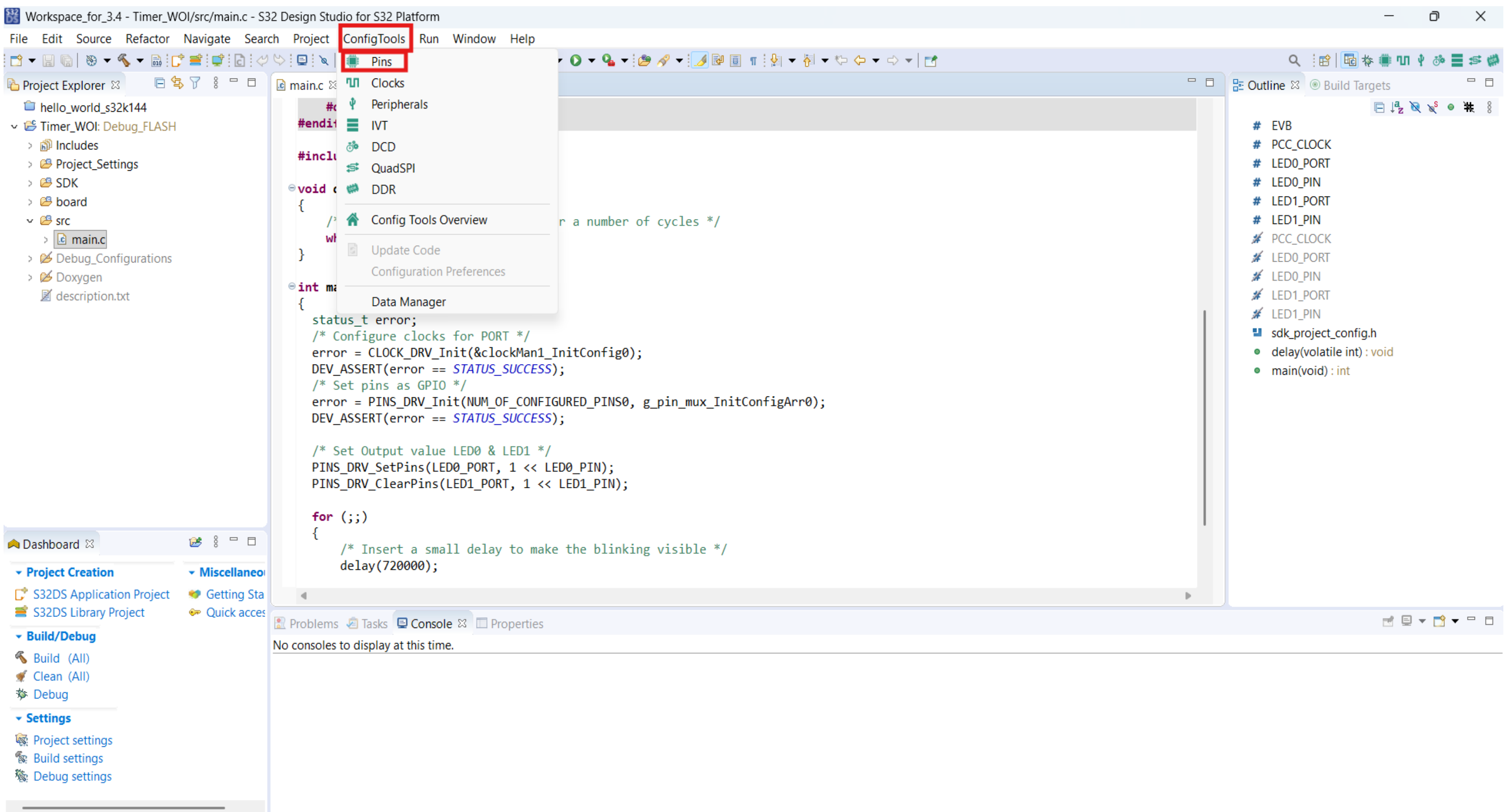

Workspace_for_3.4 - S32 Design Studio for S32 Platform
File Edit Source Refactor Navigate Search Project ConfigTools Run Window Help
Project Explorer
hello_world_s32k144: Debug_FLASH
Timer_WOI: Debug_FLASH
Outline
Build Targets
There is no active editor that provides an outline.
Close Unrelated Project
Close all projects that are not related to 'Timer_WOI'?
Always close without prompt
OK
Cancel
Dashboard
Project Creation
S32DS Application Project
S32DS Library Project
Build/Debug
Build (All)
Clean (All)
Debug
Settings
Project settings
Build settings
Debug settings
Problems
Tasks
Console
Properties
No consoles to display at this time.
Timer_WOI
337M of 721M


Workspace_for_3.4 - S32 Design Studio for S32 Platform
File Edit Source Refactor Navigate Search Project ConfigTools Run Window Help
Project Explorer
hello_world_s32k144
Timer_WOI: Debug_FLASH
Outline
Build Targets
There is no active editor that provides an outline.
Dashboard
Project Creation
S32DS Application Project
S32DS Library Project
Build/Debug
Build (All)
Clean (All)
Debug
Settings
Project settings
Build settings
Debug settings
Problems
Tasks
Console
Properties
No consoles to display at this time.
439M of 721M

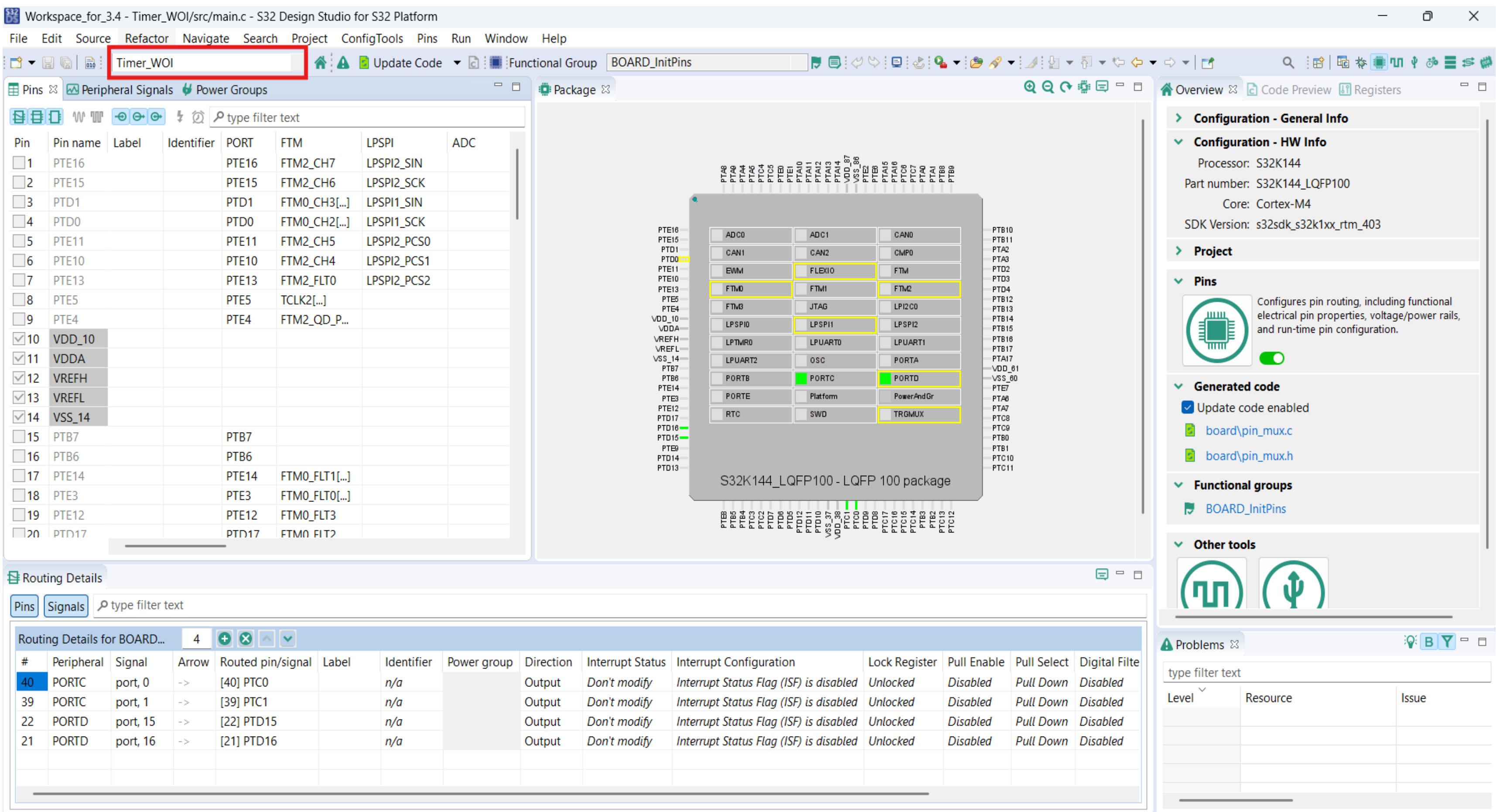
- Expand the project and open src→main.c file



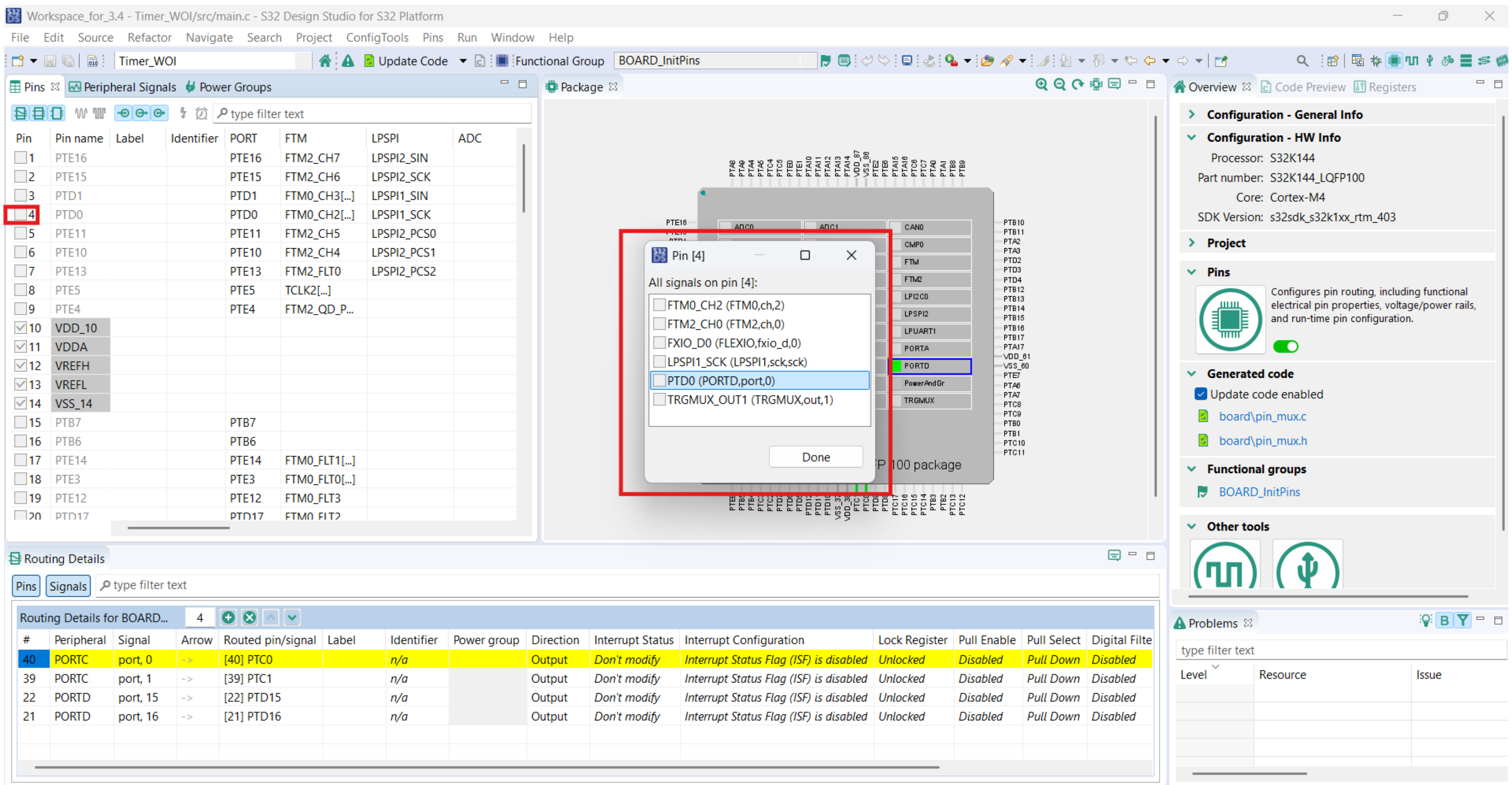
- For Pin Routing Details: Click on ConfigTools→Pins



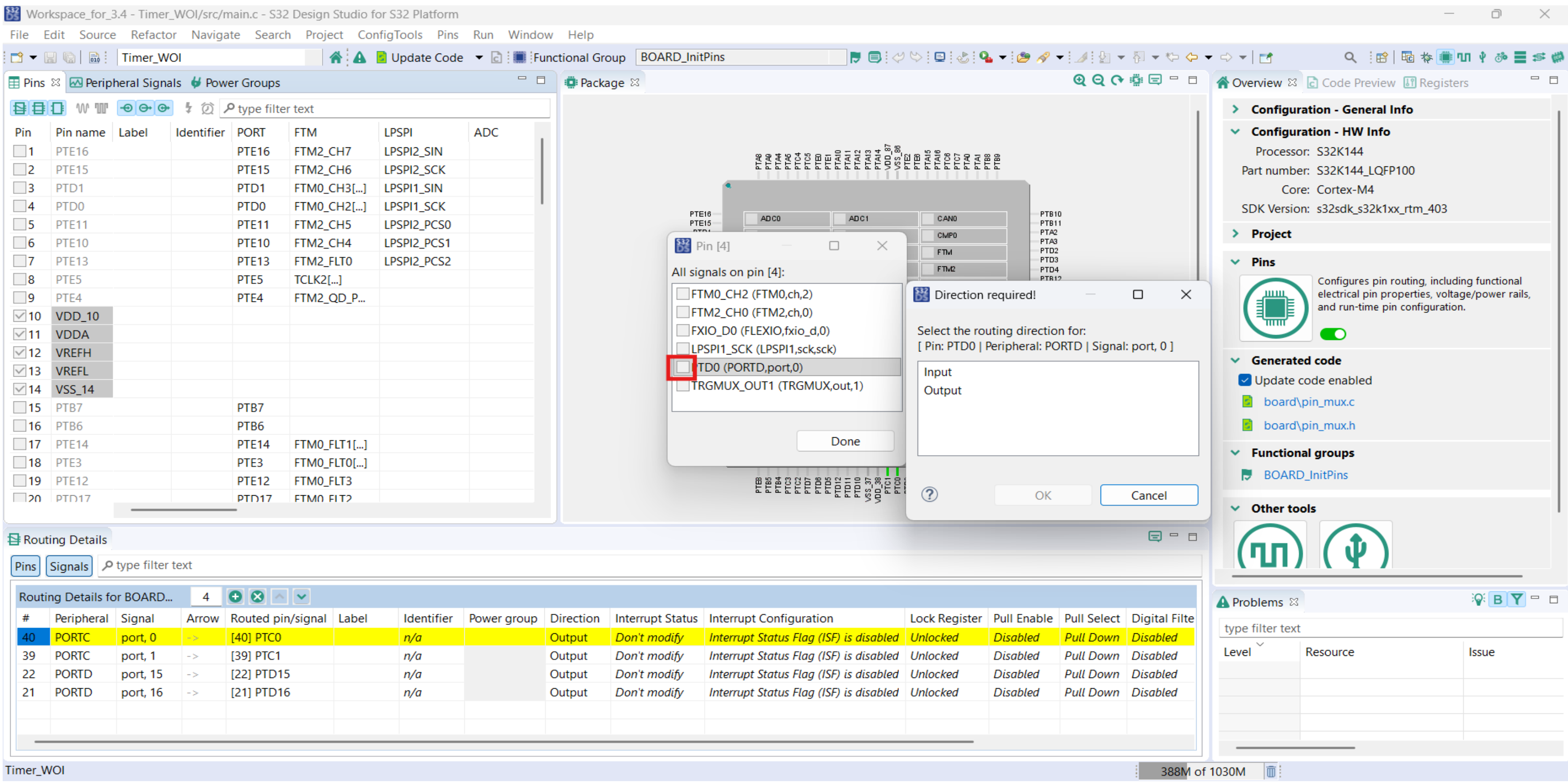
- Check your project selected



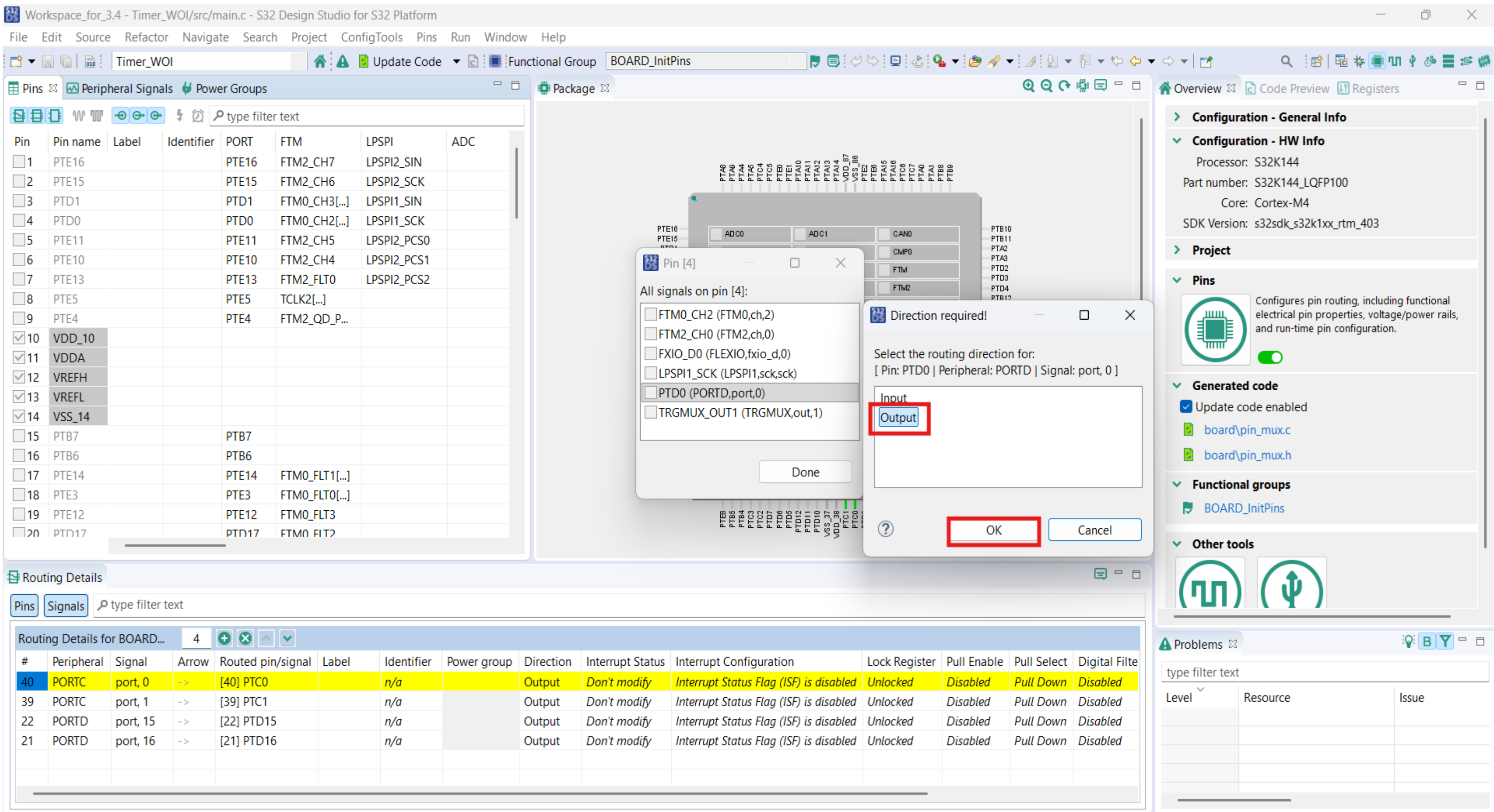
- Click on Pin 4, pop up window Pin[4] opens



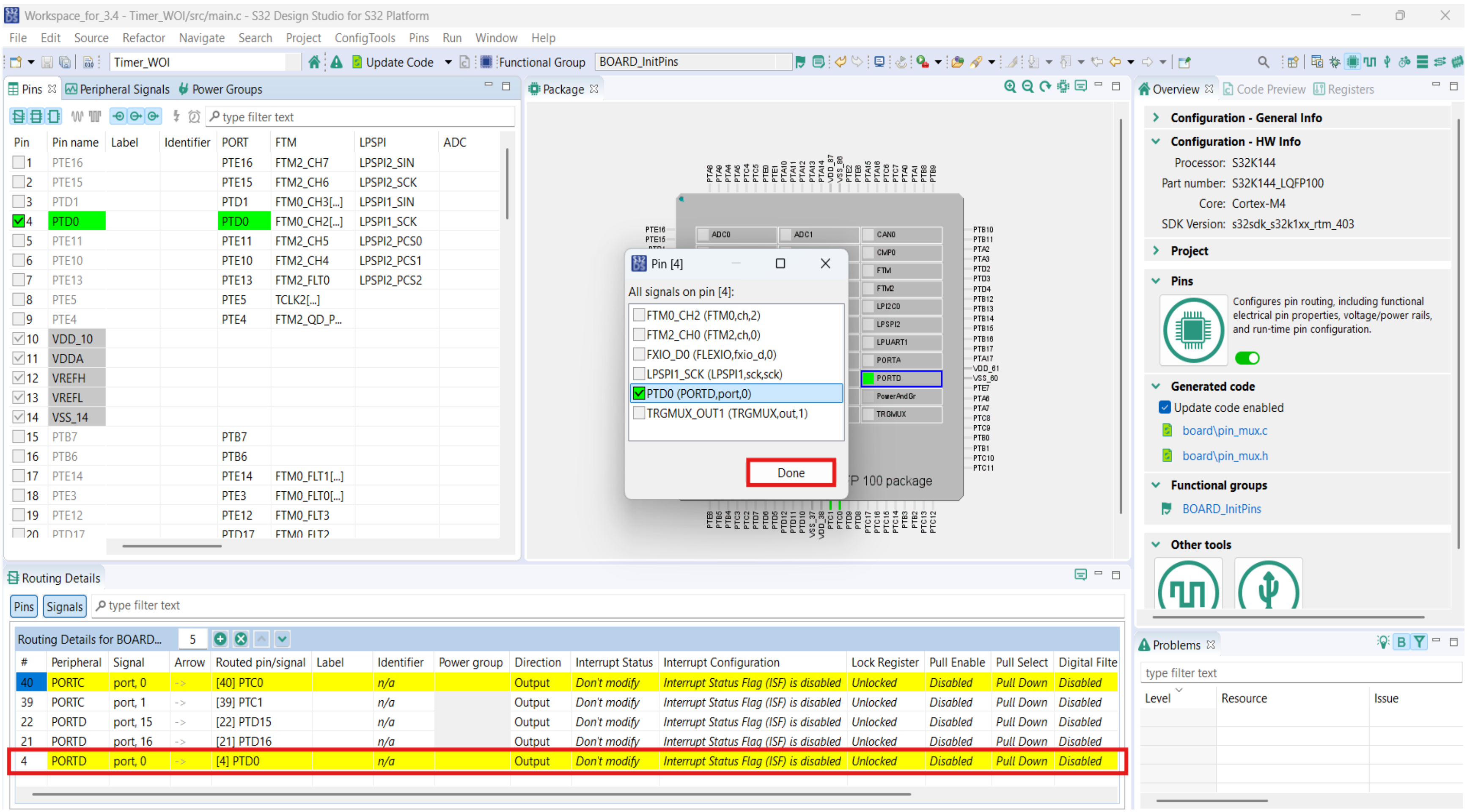
- Click on PTD0(PORTD) of Pin[4] window, pop-up Direction required window opens

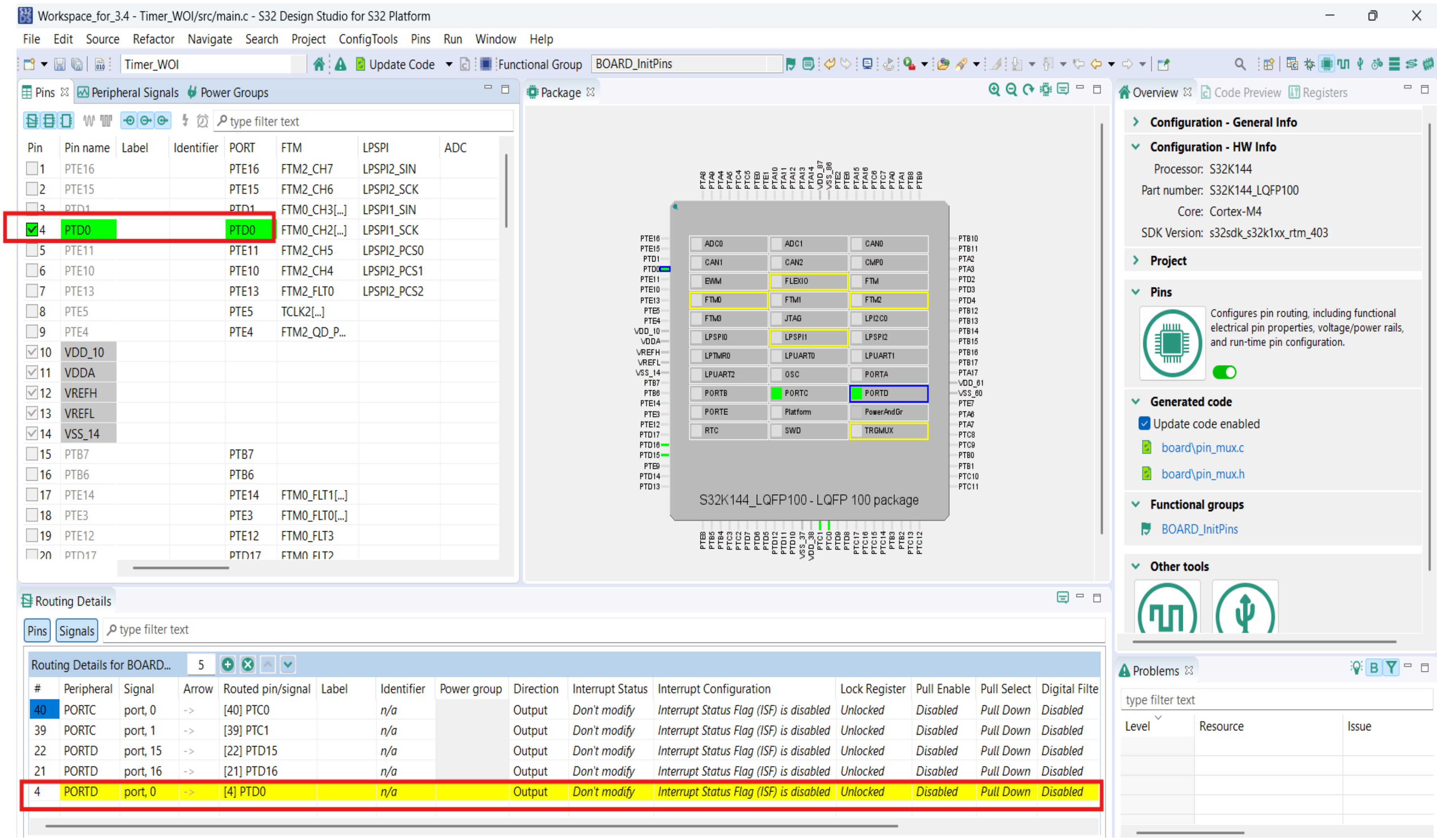


- Select Output of Direction required window and Click OK



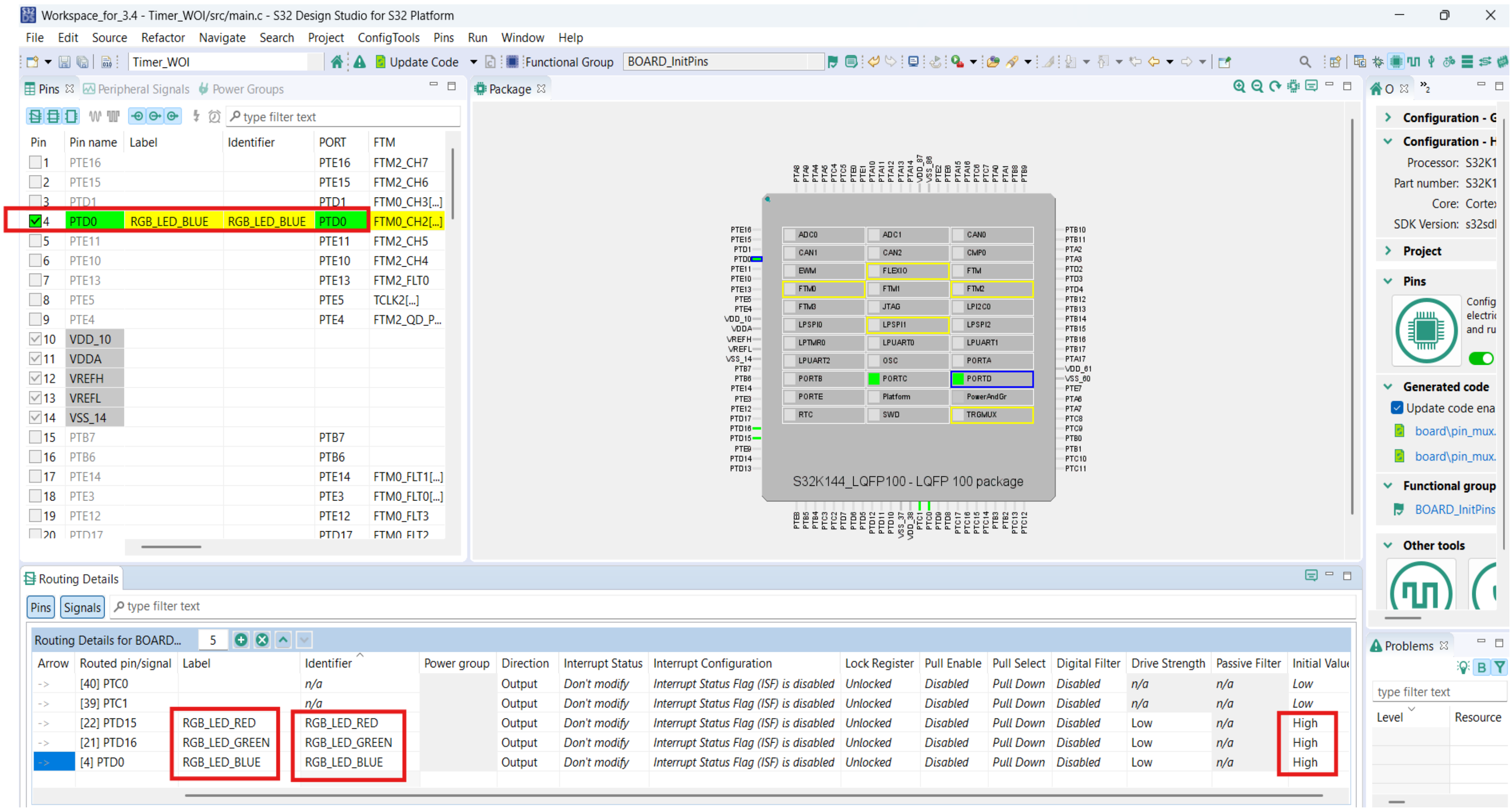
- Click on Done



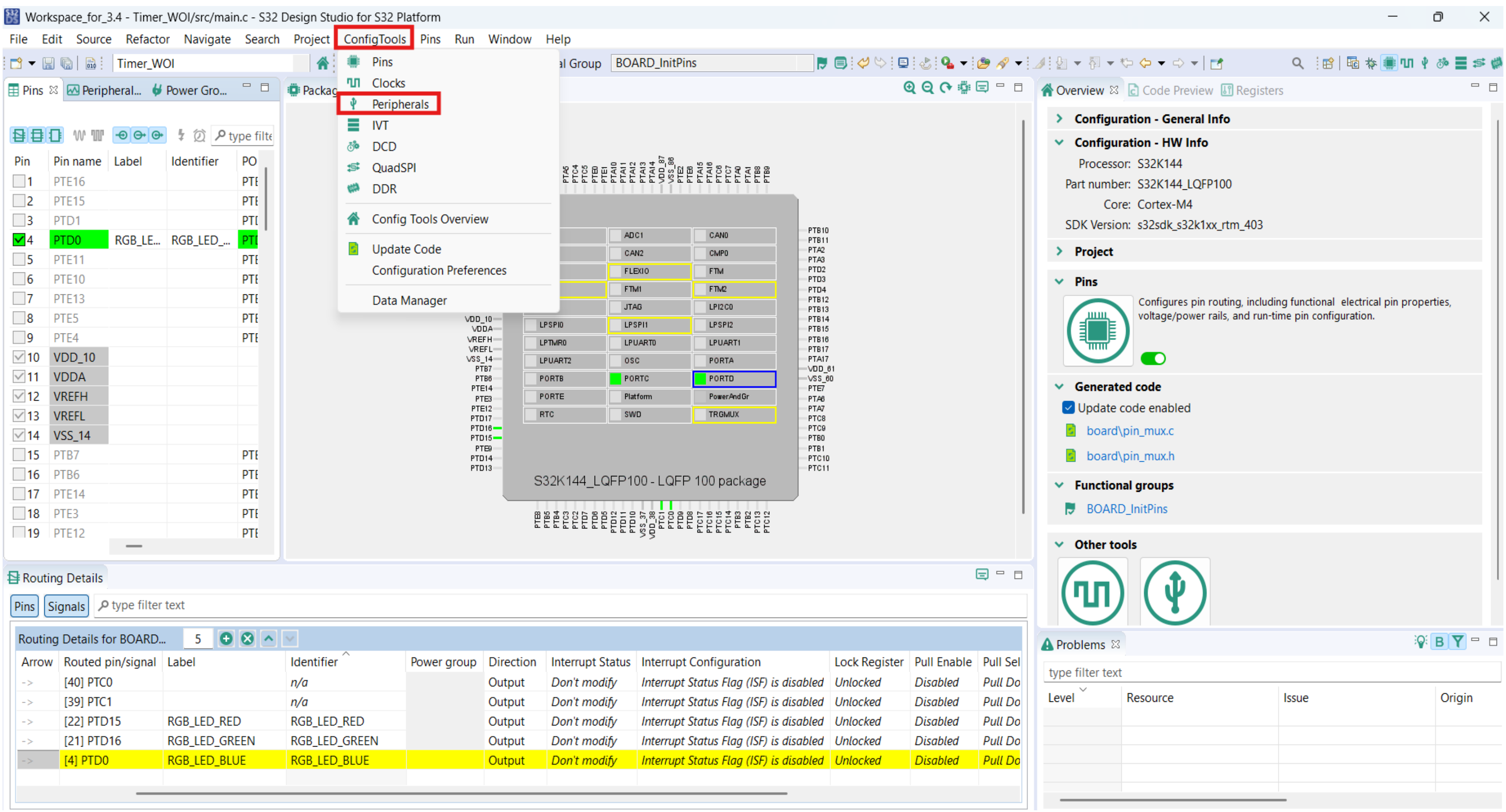


Workspace_for_3.4 - Timer_WOI/src/main.c - S32 Design Studio for S32 Platform
File Edit Source Refactor Navigate Search Project ConfigTools Pins Run Window Help
Timer_WOI
Update Code
Functional Group
BOARD_InitPins
Pins
Peripheral Signals
Power Groups
type filter text
Pin Pin name Label Identifier PORT FTM LPSPI ADC
1 PTE16 PTE16 FTM2_CH7 LPSPI2_SIN
2 PTE15 PTE15 FTM2_CH6 LPSPI2_SCK
3 PTD1 PTD1 FTM0_CH3[...] LPSPI1_SIN
4 PTD0 PTD0 FTM0_CH2[...] LPSPI1_SCK
5 PTE11 PTE11 FTM2_CH5 LPSPI2_PCS0
6 PTE10 PTE10 FTM2_CH4 LPSPI2_PCS1
7 PTE13 PTE13 FTM2_FLT0 LPSPI2_PCS2
8 PTE5 PTE5 TCLK2[...]
9 PTE4 PTE4 FTM2_QD_P...
10 VDD_10
11 VDDA
12 VREFH
13 VREFL
14 VSS_14
15 PTB7 PTB7
16 PTB6 PTB6
17 PTE14 PTE14 FTM0_FLT1[...]
18 PTE3 PTE3 FTM0_FLT0[...]
19 PTE12 PTE12 FTM0_FLT3
20 PTD17 PTD17 FTM0_FLT2
Package
S32K144_LQFP100 - LQFP 100 package
Overview
Code Preview
Registers
Configuration - General Info
Configuration - HW Info
Processor: S32K144
Part number: S32K144_LQFP100
Core: Cortex-M4
SDK Version: s32sdk_s32k1xx_rtm_403
Project
Pins
Configures pin routing, including functional electrical pin properties, voltage/power rails, and run-time pin configuration.
Generated code
Update code enabled
board\pin_mux.c
board\pin_mux.h
Functional groups
BOARD_InitPins
Other tools
Routing Details
Pins Signals
Routing Details for BOARD... 5
Peripheral Signal Arrow Routed pin/signal Label Identifier Power group Direction Interrupt Status Interrupt Configuration Lock Register Pull Enable Pull Select Digital Filte
40 PORTC port, 0 -> [40] PTC0 n/a Output Don't modify Interrupt Status Flag (ISF) is disabled Unlocked Disabled Pull Down Disabled
39 PORTC port, 1 -> [39] PTC1 n/a Output Don't modify Interrupt Status Flag (ISF) is disabled Unlocked Disabled Pull Down Disabled
22 PORTD port, 15 -> [22] PTD15 n/a Output Don't modify Interrupt Status Flag (ISF) is disabled Unlocked Disabled Pull Down Disabled
21 PORTD port, 16 -> [21] PTD16 n/a Output Don't modify Interrupt Status Flag (ISF) is disabled Unlocked Disabled Pull Down Disabled
4 PORTD port, 0 -> [4] PTD0 n/a Output Don't modify Interrupt Status Flag (ISF) is disabled Unlocked Disabled Pull Down Disabled
Problems
Level Resource Issue

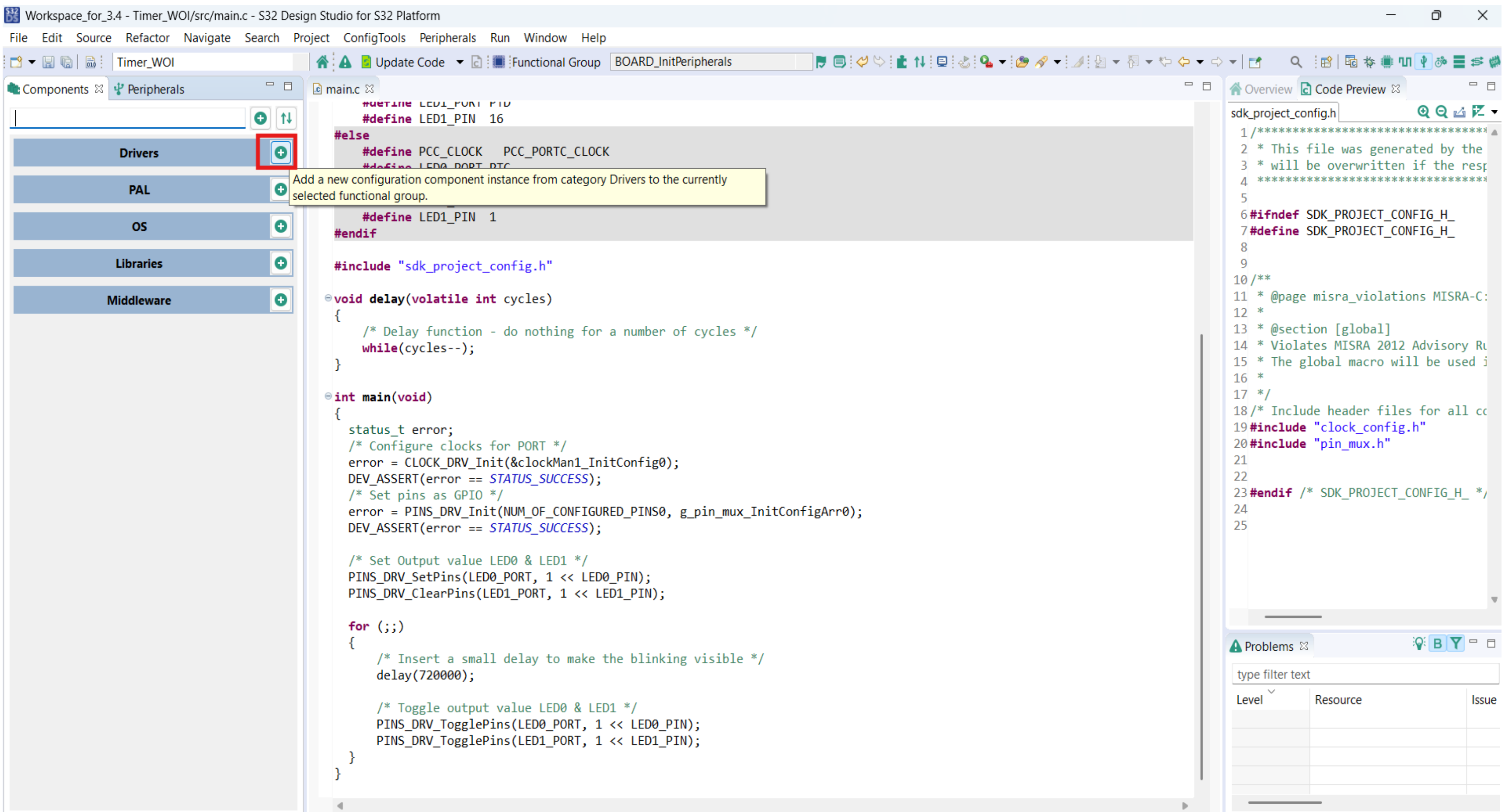
- Give Label, Identifier and Initial value as shown below:



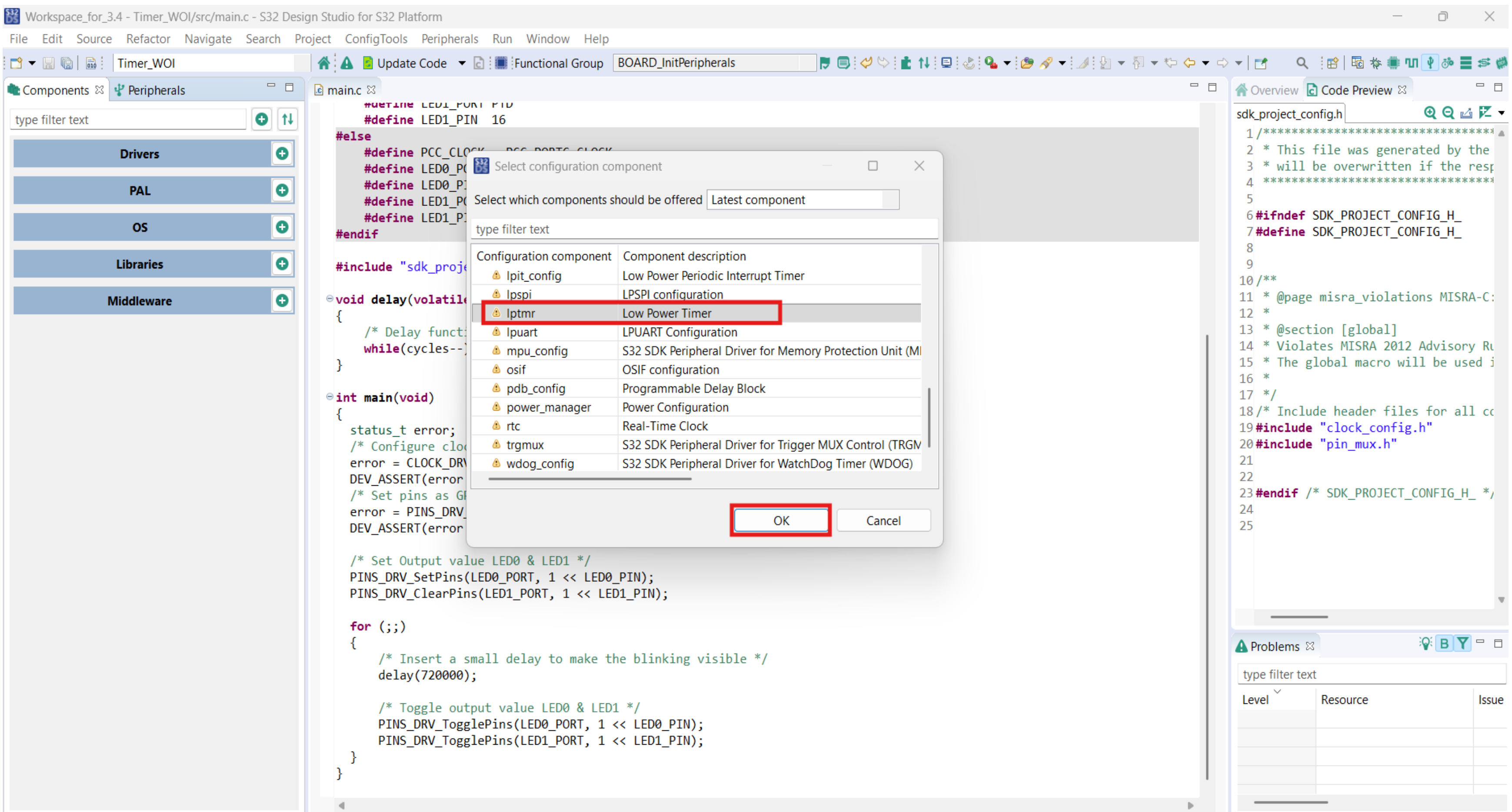
- For Peripheral Configuration: Select ConfigTools→Peripherals



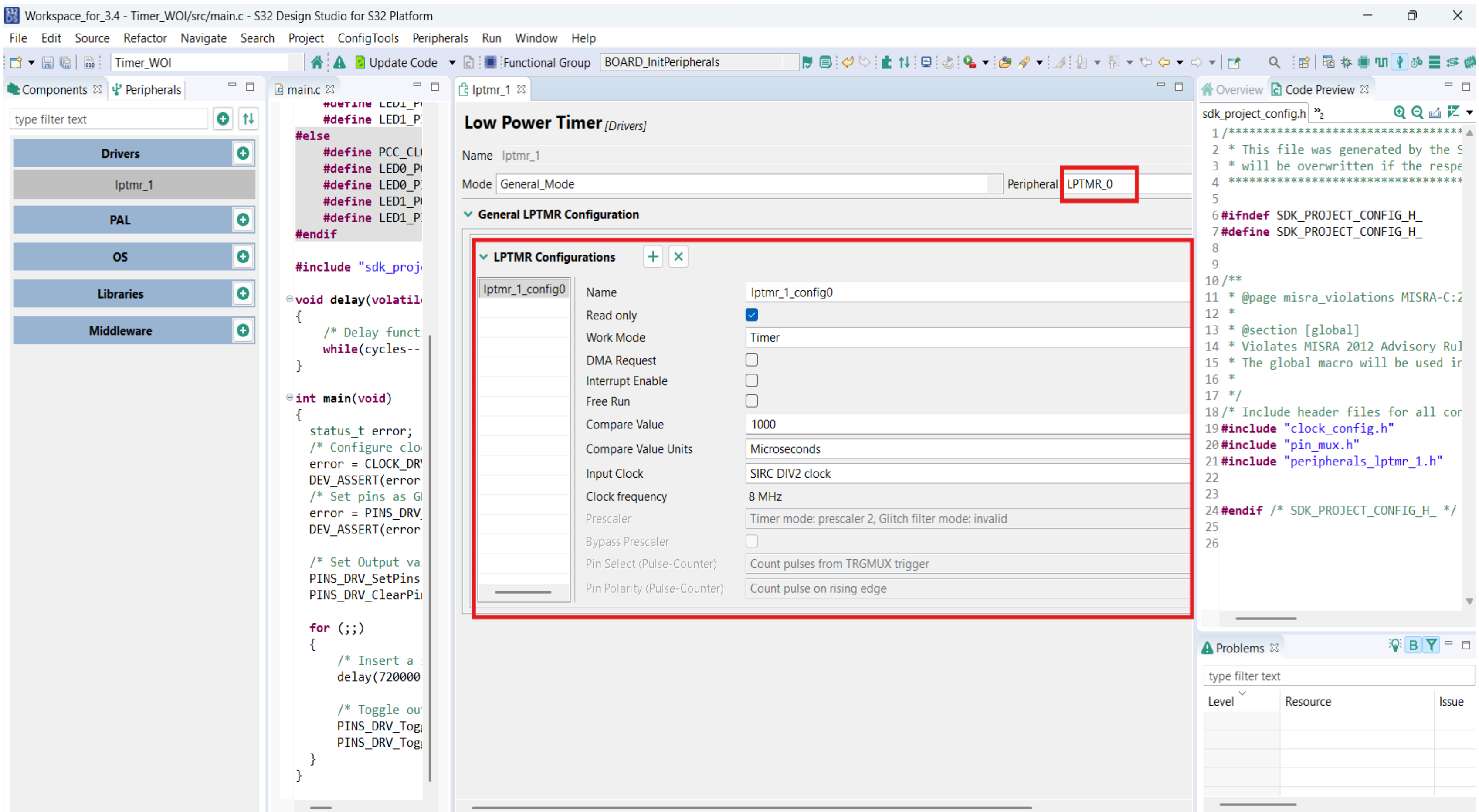
- Click on Add Drivers, new pop-up window opens



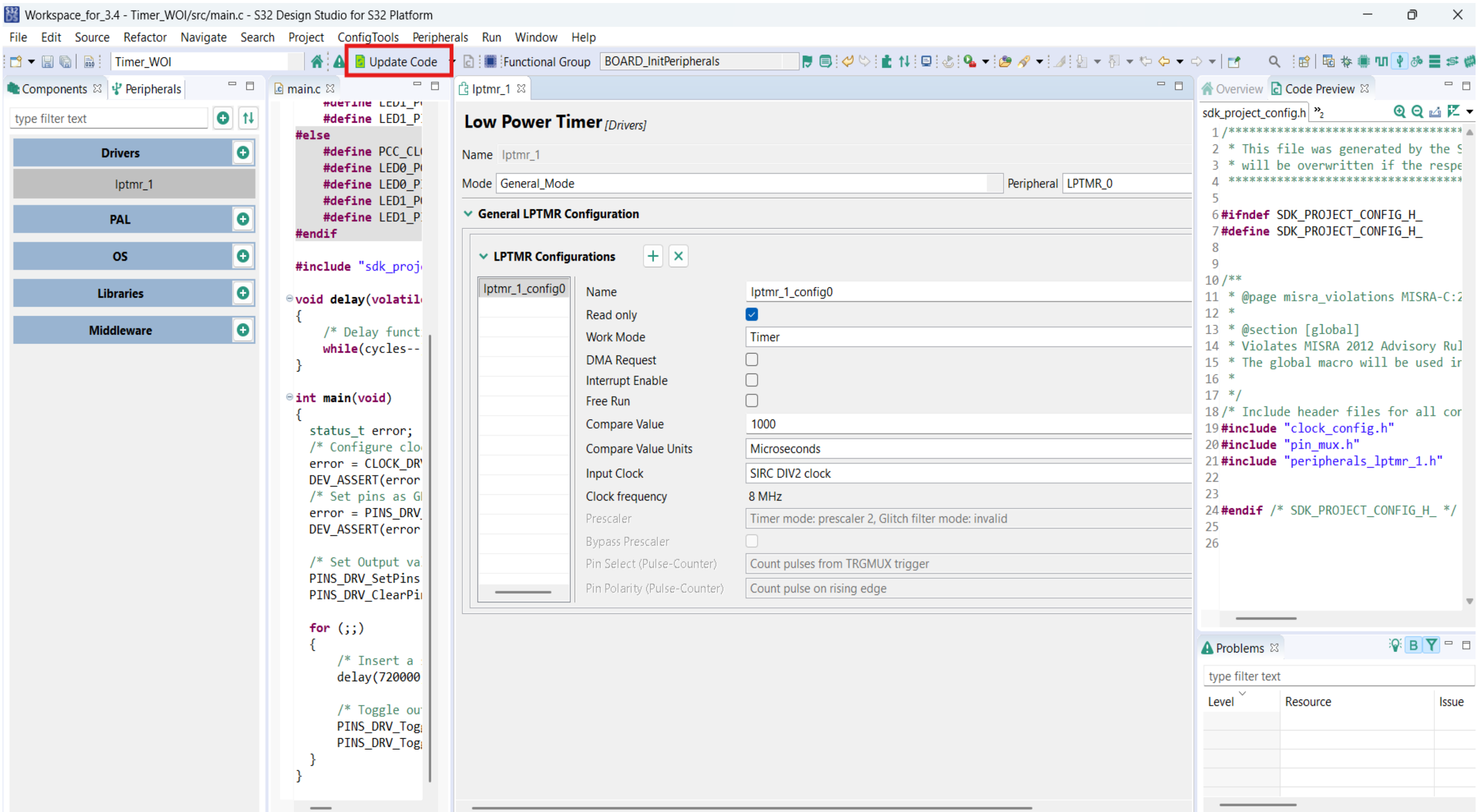
- Select **lptmr** and Click on OK



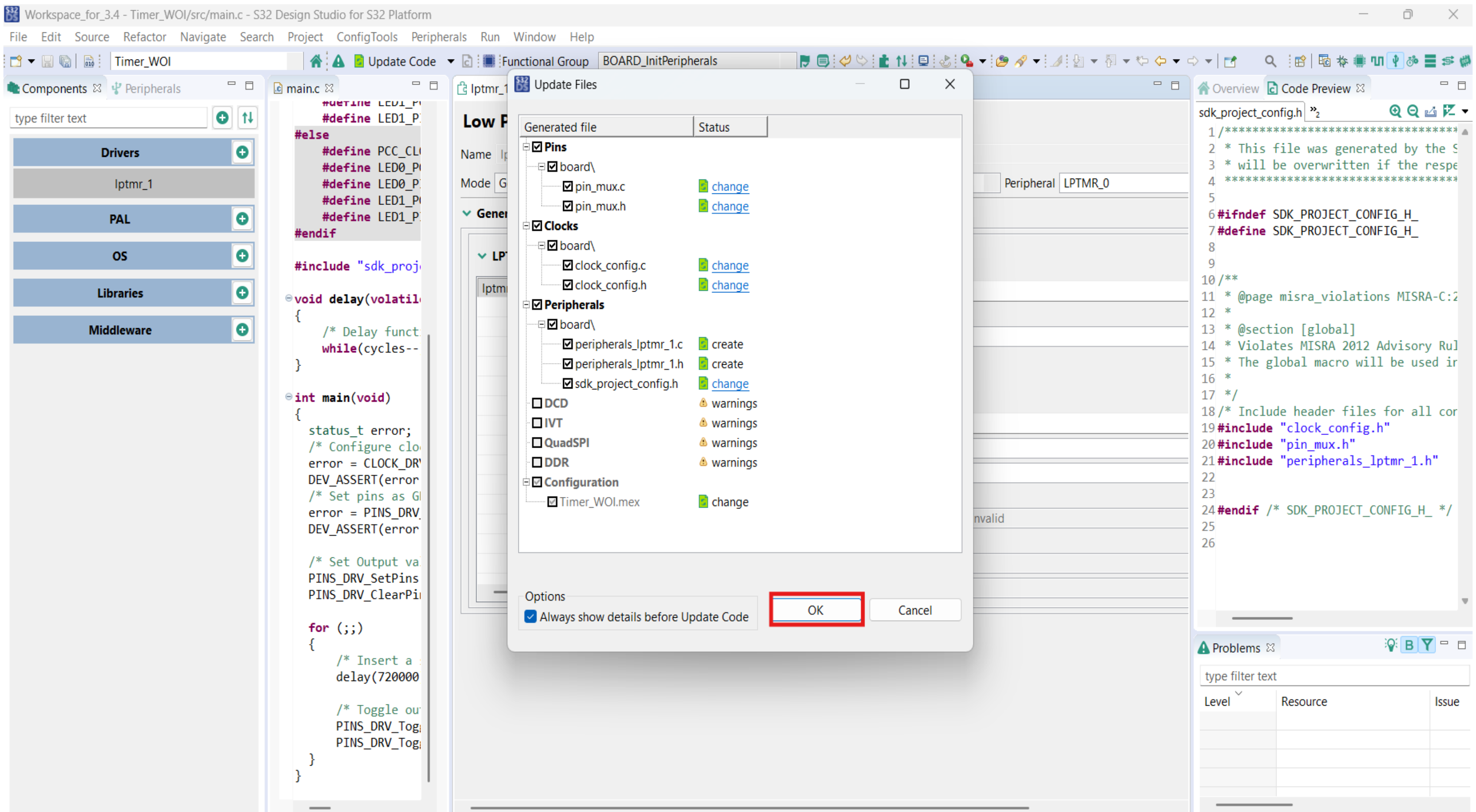
- Check settings as below for Low Power Timer



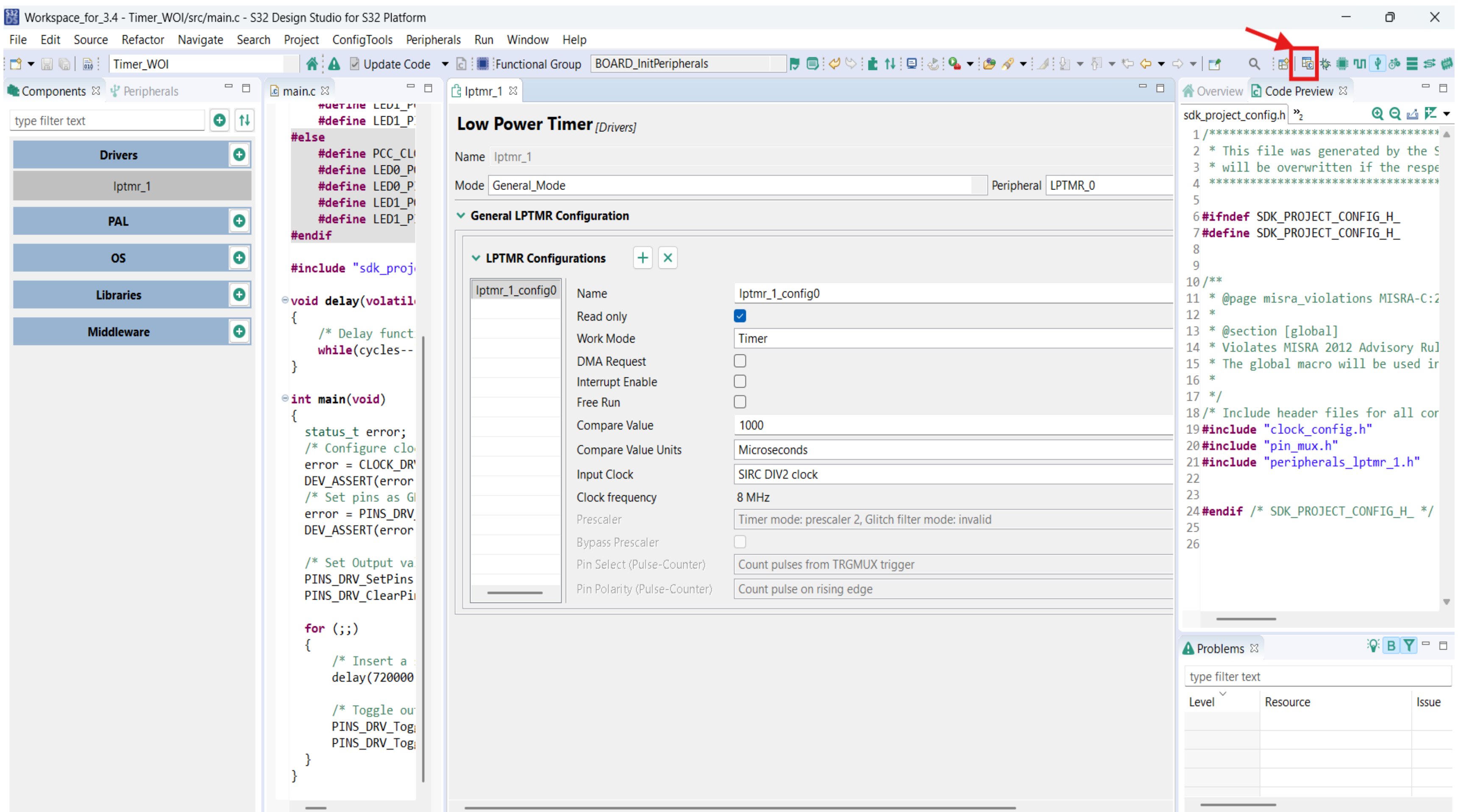
- Click on **Update Code**



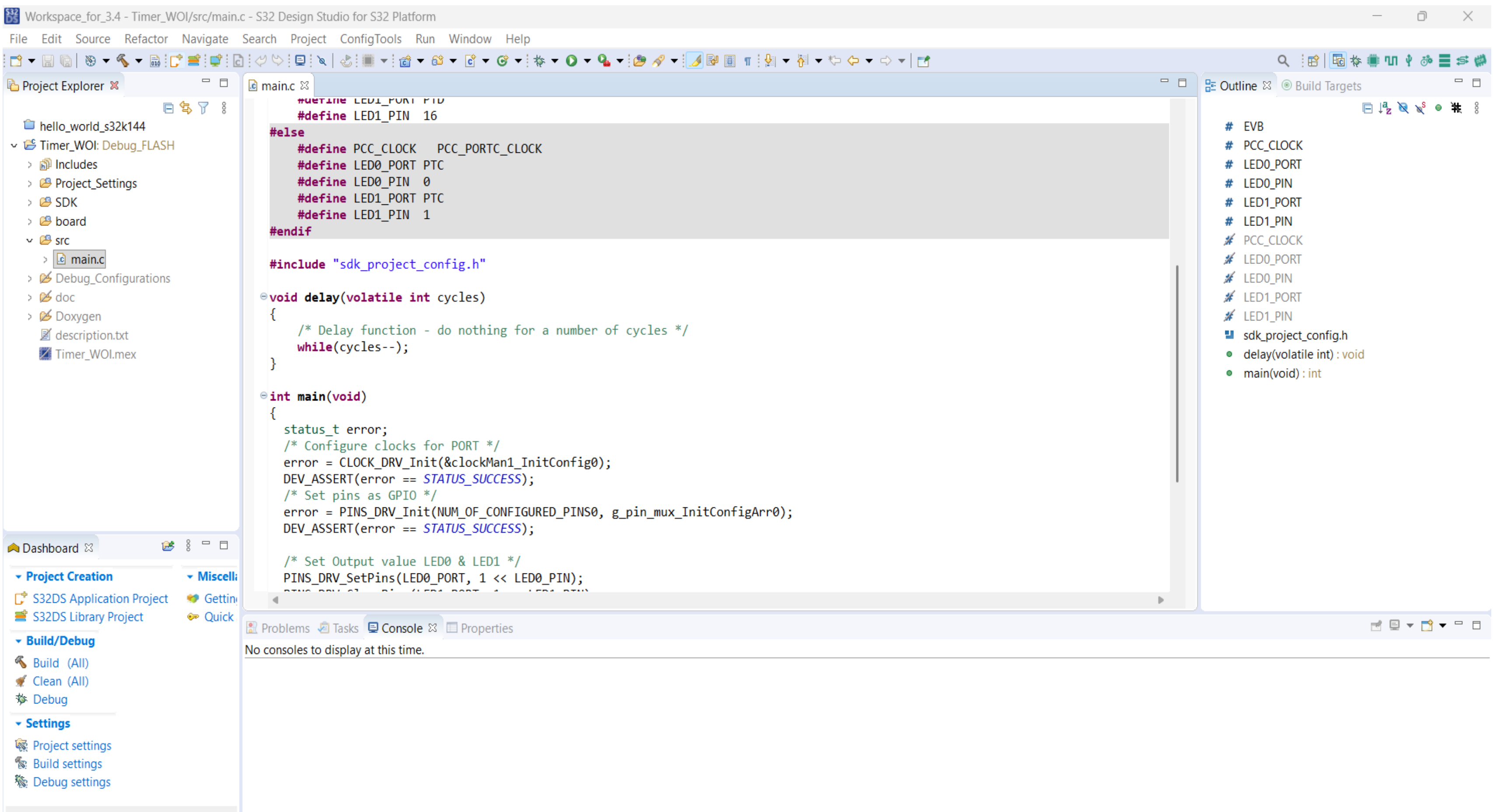
- Click on OK


Update Files
Generated file
Status
Pins
board\
pin_mux.c
change
pin_mux.h
change
Clocks
board\
clock_config.c
change
clock_config.h
change
Peripherals
board\
peripherals_lptmr_1.c
create
peripherals_lptmr_1.h
create
sdk_project_config.h
change
DCD
warnings
IVT
warnings
QuadSPI
warnings
DDR
warnings
Configuration
Timer_WOI.mex
change
Options
Always show details before Update Code
OK
Cancel

- After update click on C/C++ to change perspective



- C/C++ perspective

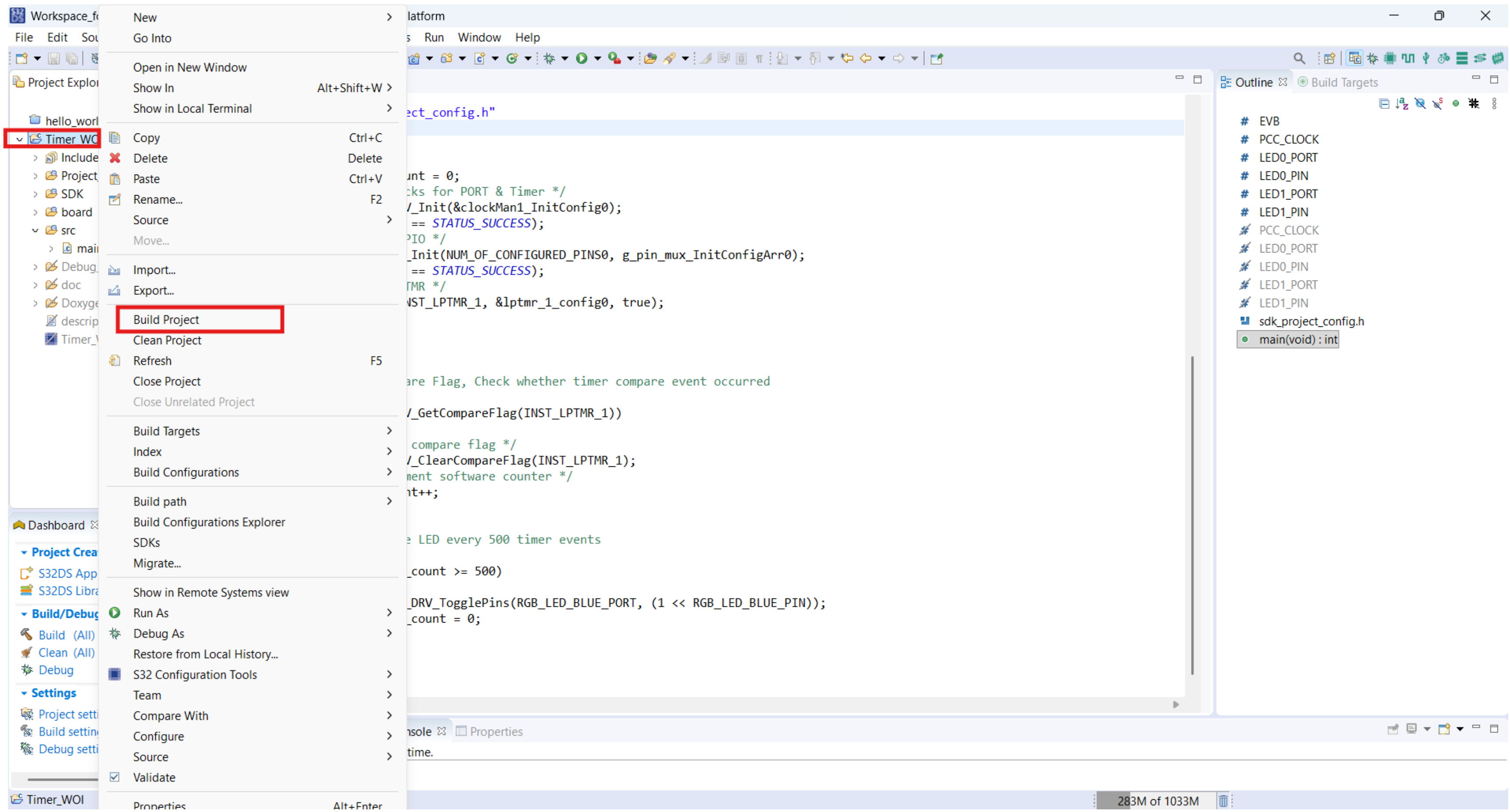


- Write C program as below and save it.

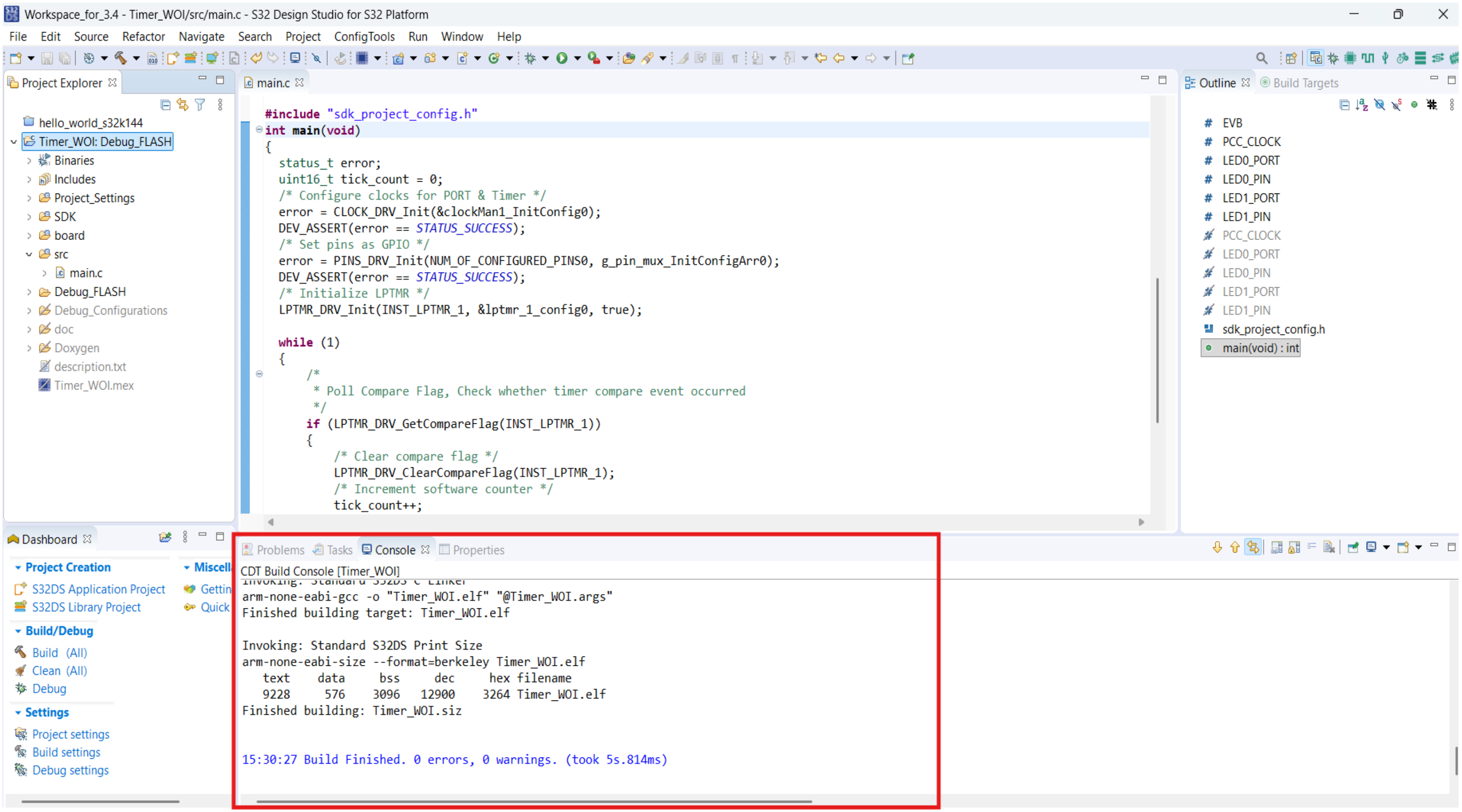
*main.c

```
#include "sdk_project_config.h"
int main(void)
{
    status_t error;
    uint16_t tick_count = 0;
    /* Configure clocks for PORT & Timer */
    error = CLOCK_DRV_Init(&clockMan1_InitConfig0);
    DEV_ASSERT(error == STATUS_SUCCESS);
    /* Set pins as GPIO */
    error = PINS_DRV_Init(NUM_OF_CONFIGURED_PINS0, g_pin_mux_InitConfigArr0);
    DEV_ASSERT(error == STATUS_SUCCESS);
    /* Initialize LPTMR */
    LPTMR_DRV_Init(INST_LPTMR_1, &lptmr_1_config0, true);

    while (1)
    {
        /*
         * Poll Compare Flag, Check whether timer compare event occurred
         */
        if (LPTMR_DRV_GetCompareFlag(INST_LPTMR_1))
        {
            /* Clear compare flag */
            LPTMR_DRV_ClearCompareFlag(INST_LPTMR_1);
            /* Increment software counter */
            tick_count++;

            /*
             * Toggle LED every 500 timer events
             */
            if (tick_count >= 500)
            {
                PINS_DRV_TogglePins(RGB_LED_BLUE_PORT, (1 << RGB_LED_BLUE_PIN));
                tick_count = 0;
            }
        }
    }
}
```

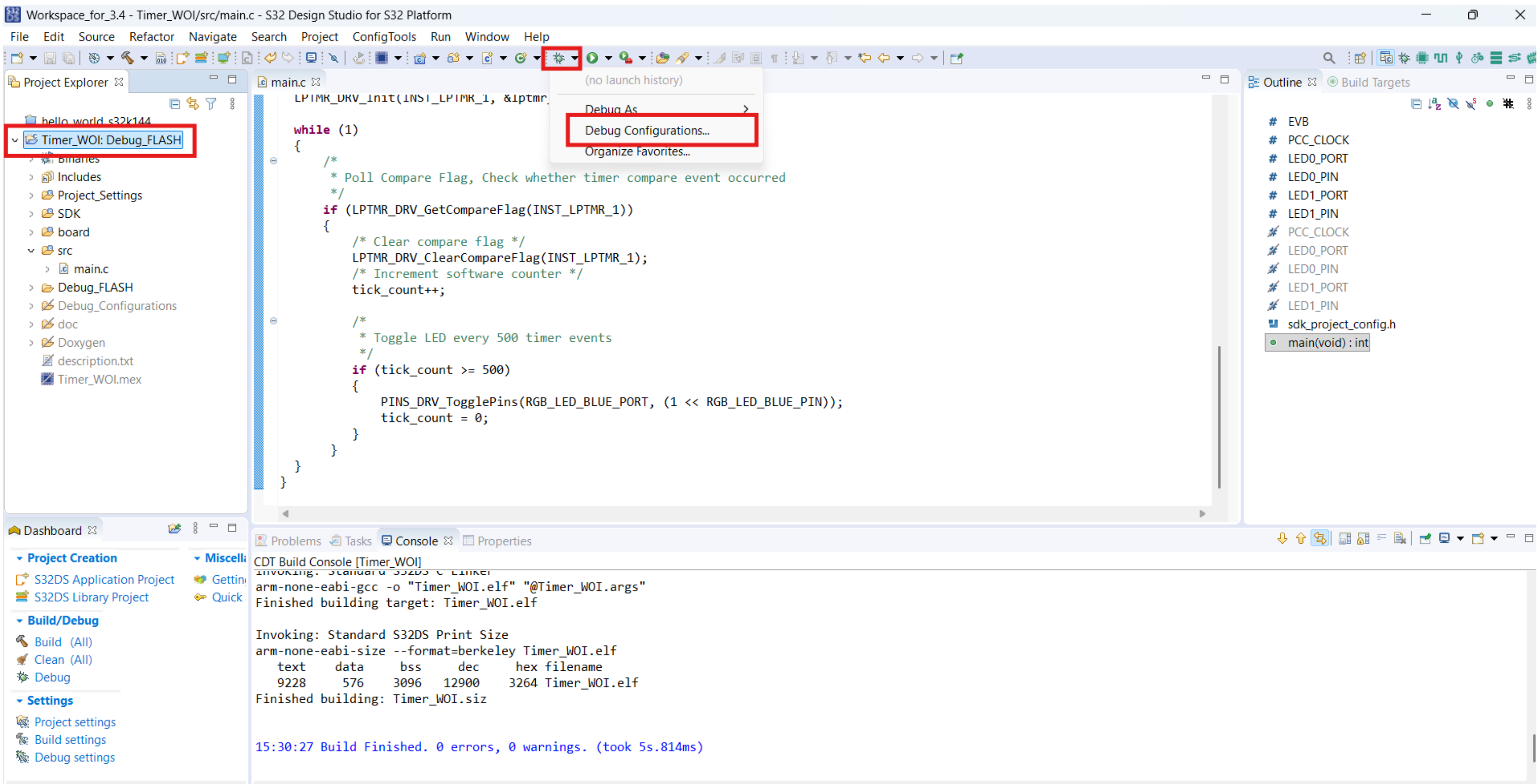
- Right click on project and click on Build Project



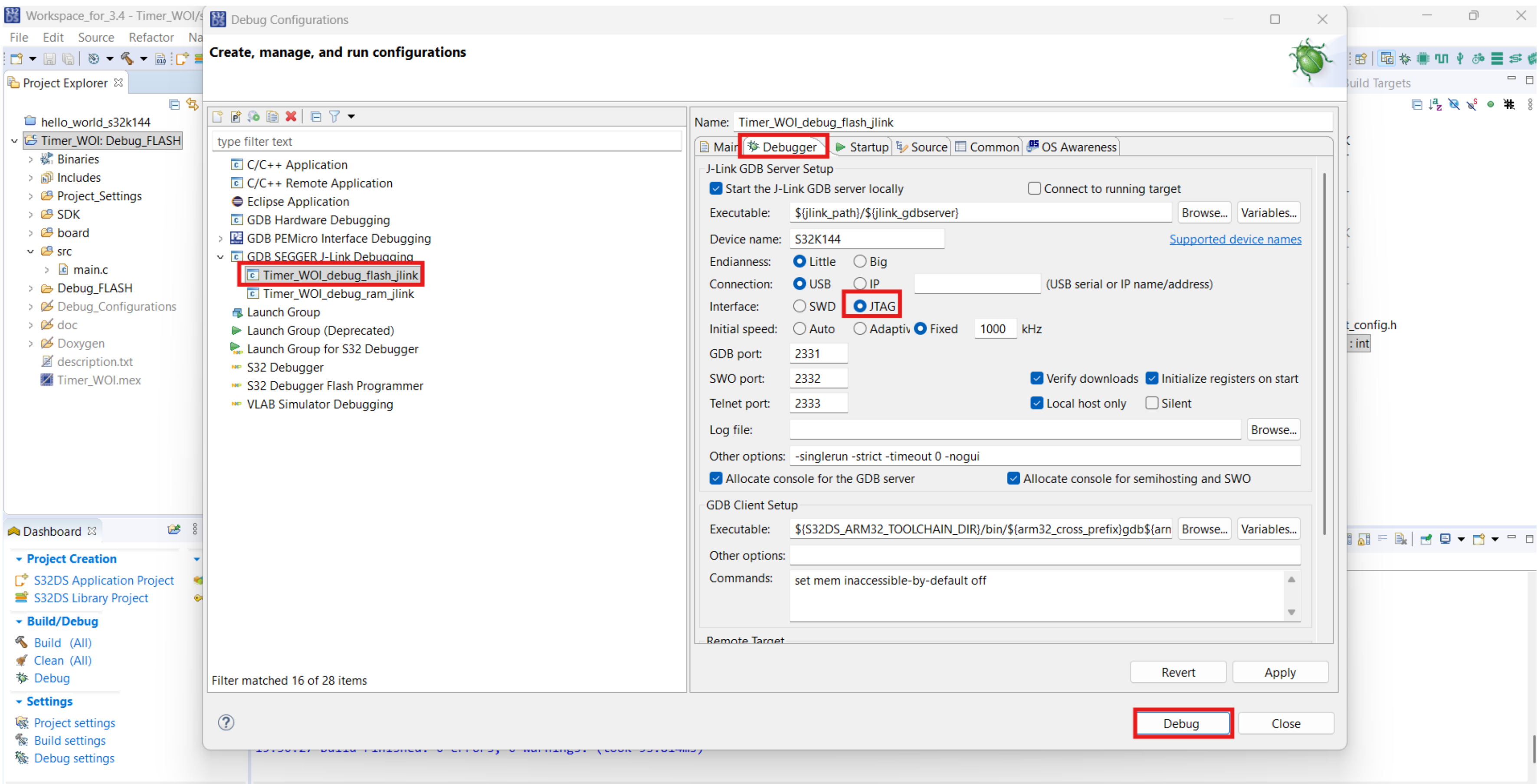
- Check for any errors



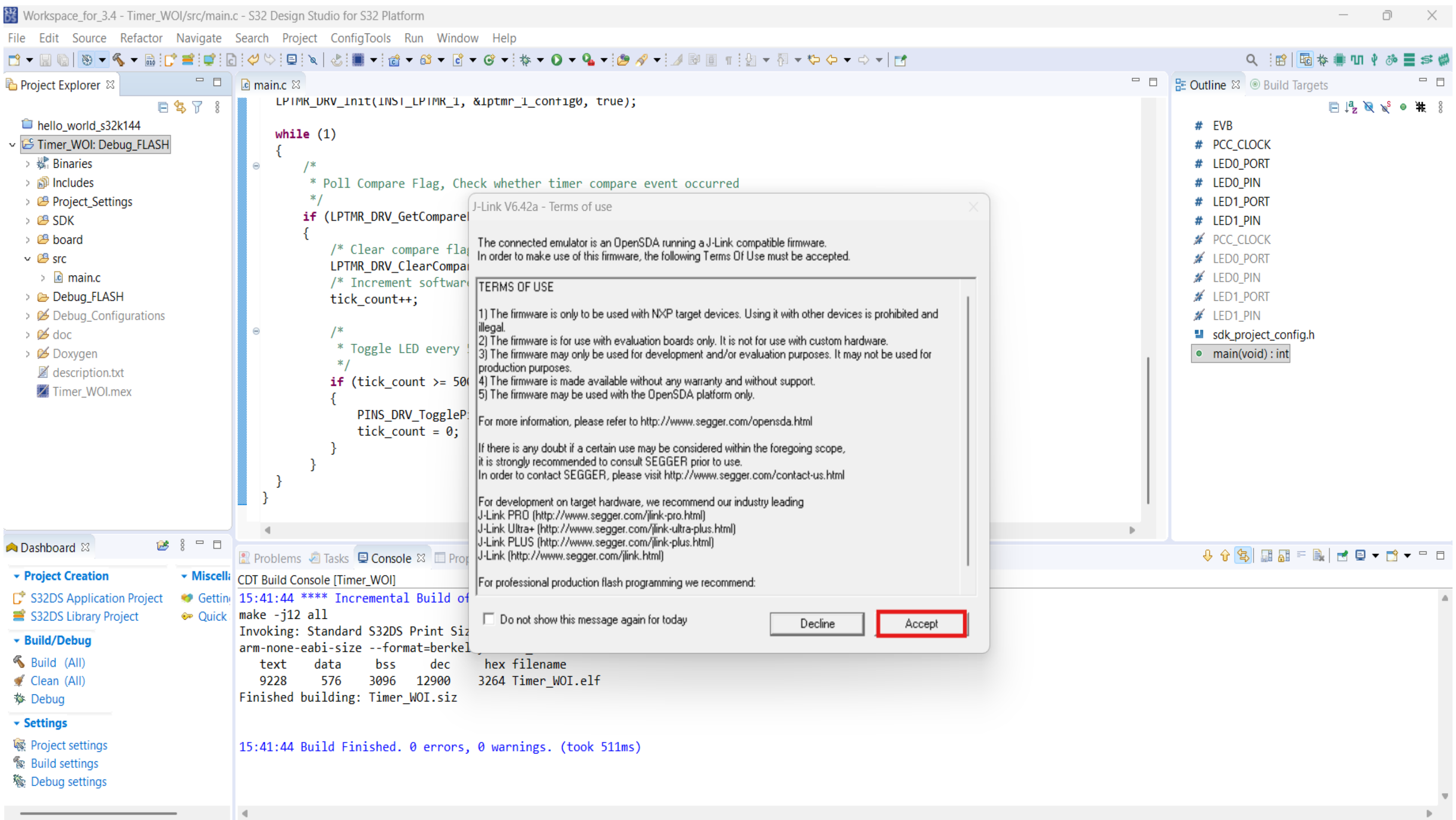
- Connect S32K144 development board to PC/Laptop
- Click on Debug→Debug Configurations



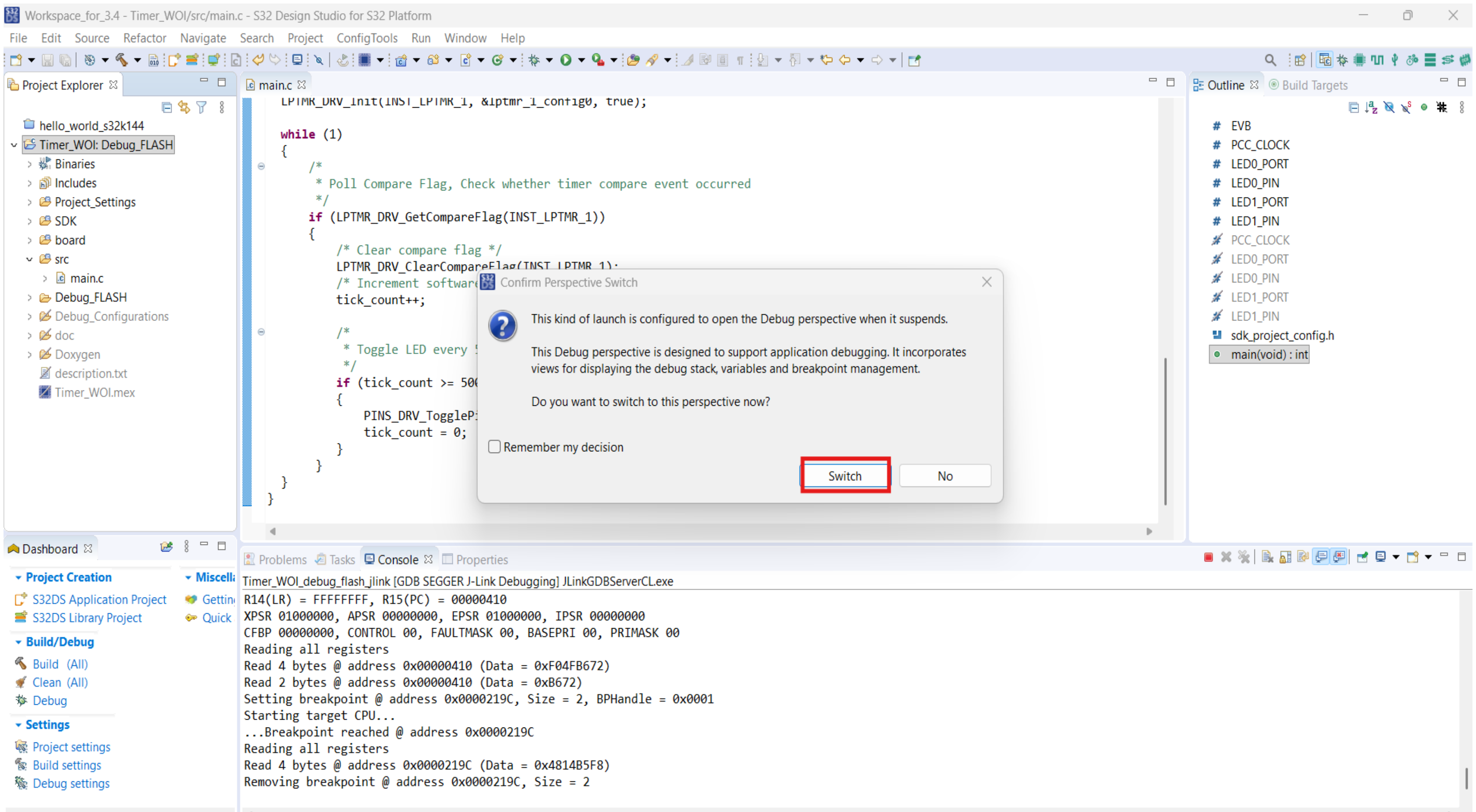
- Do configurations as below and Click on **Debug**



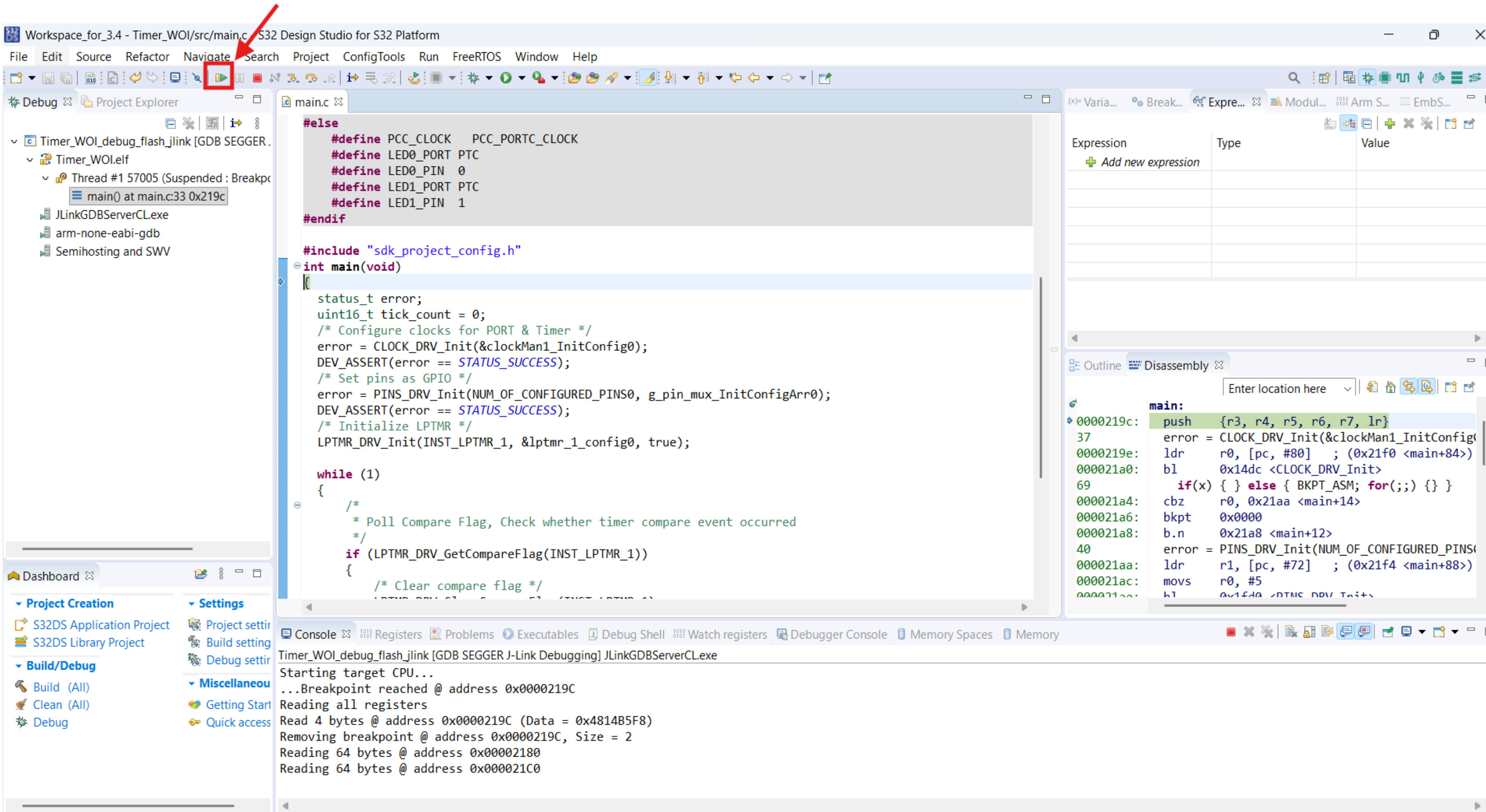
- Click on Accept



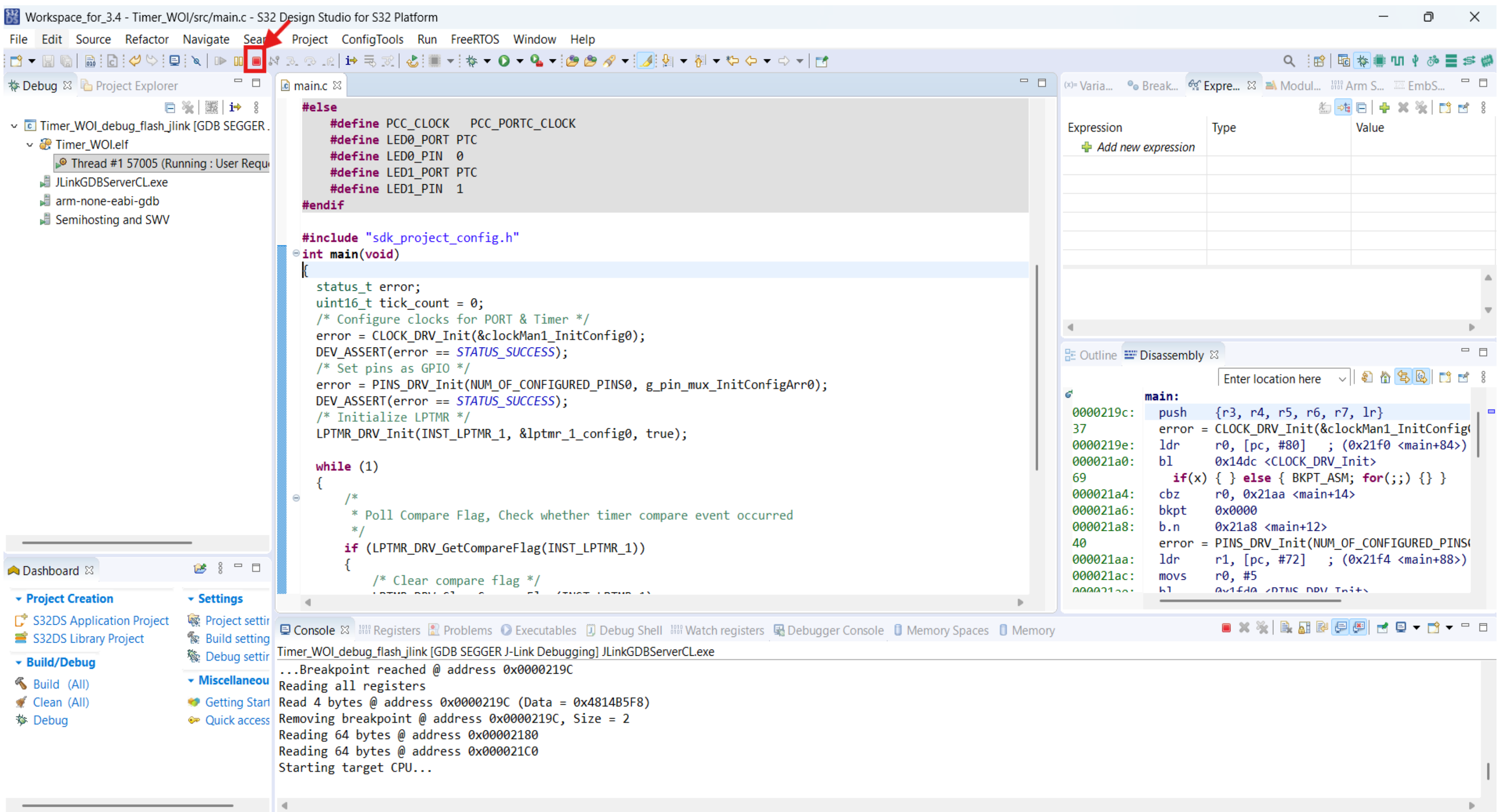
- Click on Switch, changes to Debug perspective



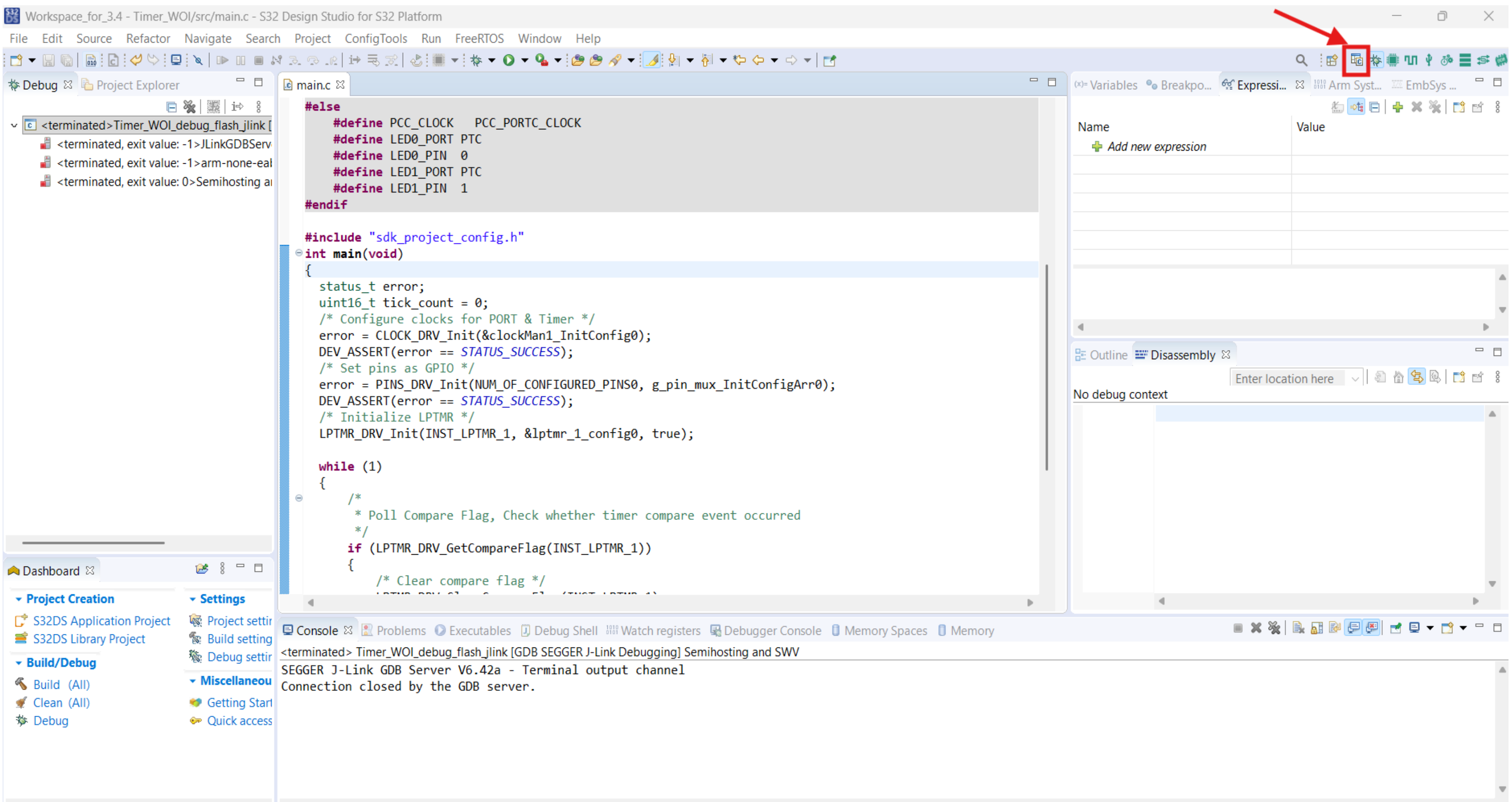
- To run code on board, click on Resume and see output on board

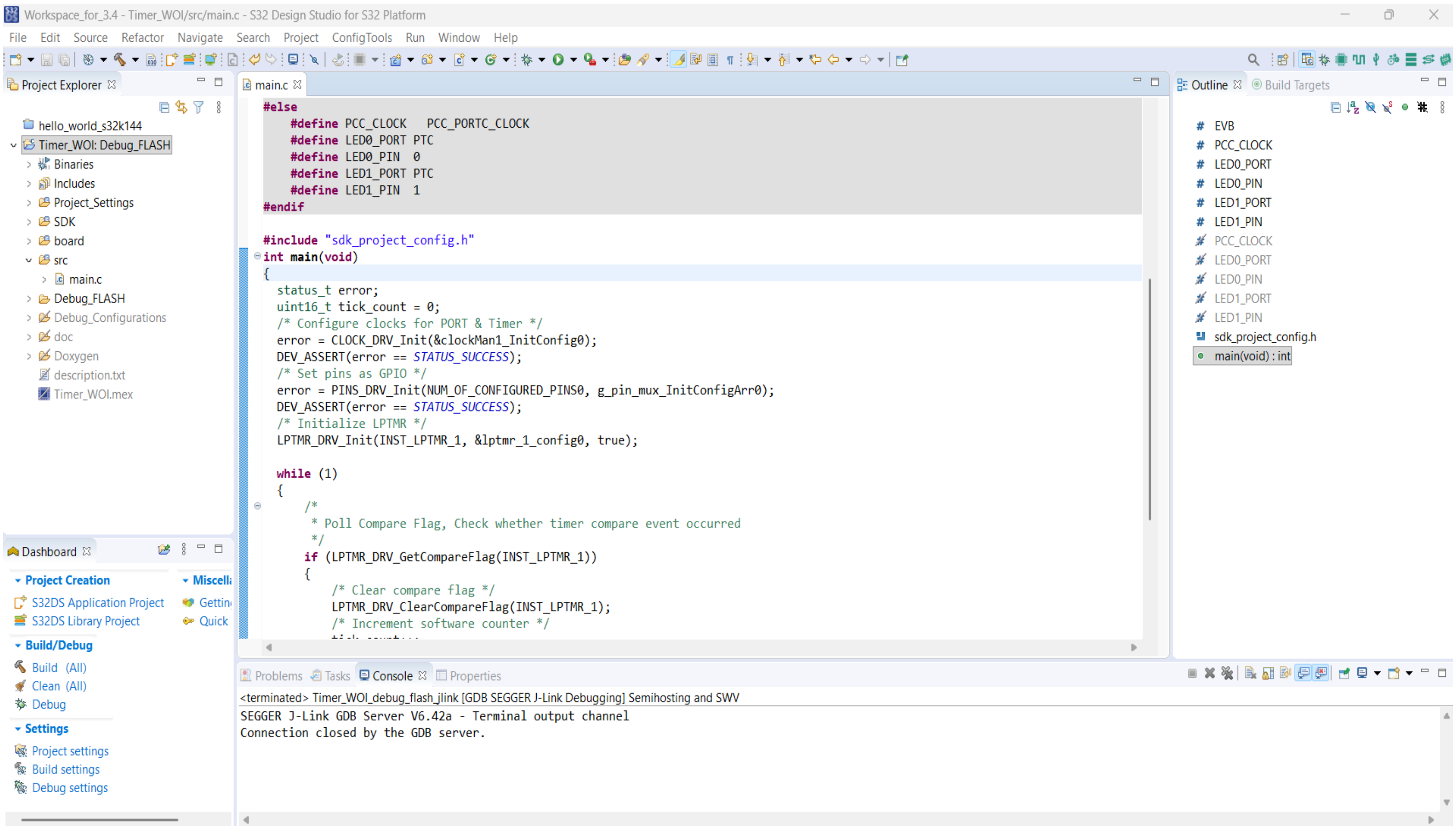


- Click on Terminate to stop debugging



- Click on C/C++ to change perspective




Workspace_for_3.4 - Timer_WOI/src/main.c - S32 Design Studio for S32 Platform
File Edit Source Refactor Navigate Search Project ConfigTools Run Window Help
Project Explorer
hello_world_s32k144
Timer_WOI: Debug_FLASH
Binaries
Includes
Project_Settings
SDK
board
src
main.c
Debug_FLASH
Debug_Configurations
doc
Doxygen
description.txt
Timer_WOI.mex
main.c
#else
#define PCC_CLOCK PCC_PORTC_CLOCK
#define LED0_PORT PTC
#define LED0_PIN 0
#define LED1_PORT PTC
#define LED1_PIN 1
#endif
#include "sdk_project_config.h"
int main(void)
{
status_t error;
uint16_t tick_count = 0;
/* Configure clocks for PORT & Timer */
error = CLOCK_DRV_Init(&clockMan1_InitConfig0);
DEV_ASSERT(error == STATUS_SUCCESS);
/* Set pins as GPIO */
error = PINS_DRV_Init(NUM_OF_CONFIGURED_PINS0, g_pin_mux_InitConfigArr0);
DEV_ASSERT(error == STATUS_SUCCESS);
/* Initialize LPTMR */
LPTMR_DRV_Init(INST_LPTMR_1, &lptmr_1_config0, true);
while (1)
{
/*
* Poll Compare Flag, Check whether timer compare event occurred
*/
if (LPTMR_DRV_GetCompareFlag(INST_LPTMR_1))
{
/* Clear compare flag */
LPTMR_DRV_ClearCompareFlag(INST_LPTMR_1);
/* Increment software counter */
Outline
Build Targets
EVB
PCC_CLOCK
LED0_PORT
LED0_PIN
LED1_PORT
LED1_PIN
PCC_CLOCK
LED0_PORT
LED0_PIN
LED1_PORT
LED1_PIN
sdk_project_config.h
main(void) : int
Dashboard
Project Creation
S32DS Application Project
S32DS Library Project
Build/Debug
Build (All)
Clean (All)
Debug
Settings
Project settings
Build settings
Debug settings
Problems Tasks Console Properties
<terminated> Timer_WOI_debug_flash_jlink [GDB SEGGER J-Link Debugging] Semihosting and SWV
SEGGER J-Link GDB Server V6.42a - Terminal output channel
Connection closed by the GDB server.

Kind Courtesy of reference books/papers/Links:

1. S32K1xx Series Reference Manual
2. S32SDK User Manual - S32K1xx RTM 4.0.3
3. https://www.nxp.com/design/design-center/software/automotive-software-and-tools/s32-design-studio-ide/s32-design-studio-for-s32-platform:S32DS-S32PLATFORM

Thank you